BEWARE OF NEGATIVE MARKING

PHYSICS

SEC-1 : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

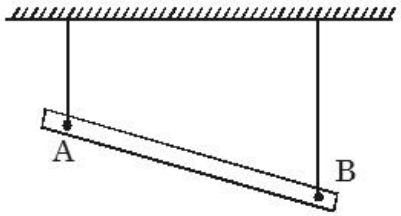
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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1. Two vertical strings having different lengths are joined to a uniform rod at points A and B. The rod is at rest. Now string connected to point B is burnt. Which of the following is/are correct just after this ?



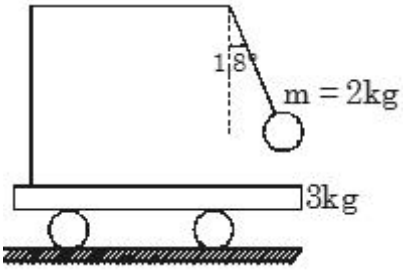
(A) Acceleration of COM is horizontal.

(B) Acceleration of COM is vertical.

(C) Acceleration of particle at A is zero

(D) Acceleration of particle at A is horizontal

2. A simple pendulum of length ℓ = 30 cm and mass m = 2kg is attached to a cart of mass 3 kg. The cart can move frictionlessly on level ground. The bob is released from rest when string makes an angle of 1.8° with the horizontal :-



(A) The amplitude of oscillation of cart is $\frac{\pi}{5}$cm.

(B) The time period of oscillation is $\frac{\pi}{5}$sec

(C) The cart and bob oscillate with phase difference of π.

(D) The time period of oscillation is $\frac{\pi\sqrt{3}}{5}$sec.

3. Consider the potential energy given by

$U = \frac{x^4}{4} - \frac{x^3}{3} - 3x^2$

(A) x = 0 is position of stable equilibrium.

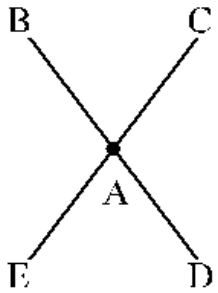
(B) x = 3 is position of stable equilibrium.

(C) If body starts from x = –2, it can go to x = 3 if its kinetic energy is greater than $\frac{16}{3}$J.

(D) If body starts from x = –2, it can go to x = 3 even if its initial kinetic energy is zero.

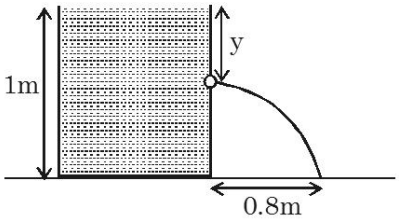
Space for Rough Work

4. Four identical rods which have thermally insulated lateral surfaces are joined at point A. Points B,C,D & E are connected to large reservoirs. If heat flows into the junction from point B at rate of 1 W and from point C at 3 W inside, flows out from D at 5 W, which relation(s) is/are correct for temperature of these points ?



(A) $T_A < T_E$ (B) $T_B = T_C$ (C) $T_C > T_D$ (D) $T_B = T_E$

5. In a tank filled with water, a small hole is made on the side. The emerging water jet falls to a distance of 0.8 m from the base as shown (y is distance of hole from above) :-



(A) As the level of water falls, the range decreases.

(B) As the level of water falls, the time of flight for water jet remains the same.

(C) y can be 0.8 m.

(D) y can be 0.2 m.

Space for Rough Work

6. In a standard YDSE setup, the fringe width on the screen is 1.5 mm. When a thin glass film is pasted in front of the upper slit, the fringe pattern shifts up. But it is seen that at a point P above central maxima where intensity was one fourth the intensity at central maxima, intensity remains the same. There were no maxima between central maxima and point P before film was introduced. What can be the thickness of the film ? Take $\mu = 1.5$, $\lambda = 450$ nm :-

(A) 3×10^{-7}m (B) 5×10^{-7} m (C) 9×10^{-7} m (D) 1.4×10^{-6} m

Space for Rough Work

SEC-2 : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks : –1 In all other cases

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1. A charged particle having non zero velocity is subjected to certain conditions given in List-I, List-II gives possible trajectories of the particle. Match the conditions in List-I with the results in List-II. Given all the fields are non-zero.

	List-I		List-II
(P)	In only uniform electric field	(1)	The path of the charged particle may be a straight line
(Q)	In only uniform magnetic field	(2)	The path of the charged particle may be a parabola
(R)	In uniform magnetic and uniform electric field	(3)	The path of the charged particle may be circular
(S)	Subjected to a net force of constant magnitude	(4)	The path of the charged particle may be a helical with uniform or non-uniform pitch.
		(5)	None of these

(A) P → 1,2;Q → 1,3,4;R → 1,4;S → 1,2,3,4

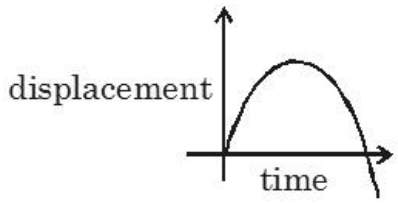
(B) P → 1,2,3;Q → 1,3,4;R → 1,4;S → 1,2,3

(C) P → 1,2;Q → 1,2,3,4;R → 1,3,4;S → 1,2,3,4

(D) P → 1,2,3;Q → 1,3;R → 1,3,4;S → 1,2,3

Space for Rough Work

2. In the list–I, there are certain situations depicted and list–II gives the description of speed and acceleration of particle. Match them. Neglect air resistance and assume uniform gravitational field wherever applicable. Consider time before it hits the ground.

	List-I		List-II
(P)	 Particle moving in 1 dimension	(1)	speed increases & then decreases
(Q)	$x = t^2 + 3t$	(2)	speed decreases & then increases
(R)	A projectile thrown on level ground at an angle to the horizontal	(3)	speed increases continuously.
(S)	A ball thrown vertically down from a cliff.	(4)	speed decreases continuously
		(5)	acceleration is initially opposite to velocity.
		(6)	acceleration is initially in the same direction as the velocity

(A) P → 1,5;Q → 2,6;R → 4;S → 2,5

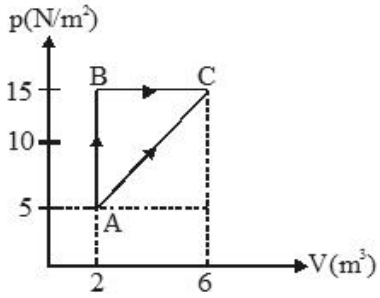
(B) P → 2,5;Q → 3,6;R → 2;S → 3,6

(C) P → 1,5;Q → 4,6;R → 1,6;S → 2,5

(D) P → 2,6;Q → 2;R → 3,6;S → 3,6

Space for Rough Work

3. An ideal gas is taken along the reversible processes as represented by the adjoining diagram.



	List-I		List-II
(P)	For process B → C	(1)	$\Delta Q > 0$
(Q)	For process A → B	(2)	$\Delta W > 0$
(R)	For cycle A → B → C → A	(3)	$\Delta U > 0$
(S)	For process C → A	(4)	$\Delta W = 0$
		(5)	$\Delta Q < 0$

(A) P → 1,2,3;Q → 1,2,5;R → 1,2,3;S → 1,2

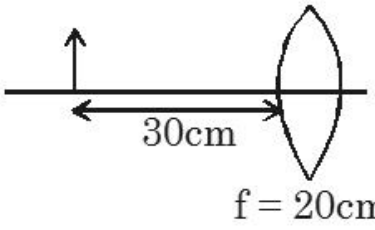
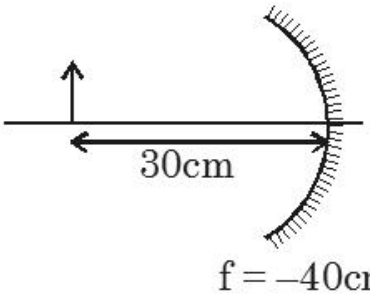
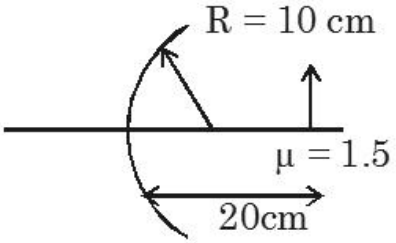
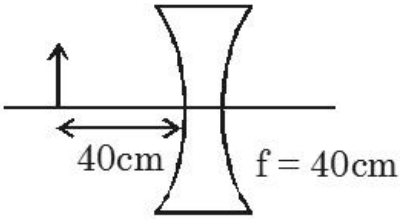
(B) P → 1,2;Q → 1,3,4;R → 1,4,5;S → 5

(C) P → 1,2,3;Q → 1,3,4;R → 1,2;S → 5

(D) P → 1,3;Q → 1,3,4;R → 5;S → 4,5

Space for Rough Work

4. Match the list–1 with nature of image in list–II (Consider all optical devices are kept in air)

	List-I		List-II
(P)		(1)	Erect image
(Q)		(2)	Inverted image
(R)		(3)	magnified image
(S)		(4)	diminished image
		(5)	real image
		(6)	virtual image

(A) P → 2,3,5;Q → 1,4,6;R → 1,3,6;S → 2,3,5

(B) P → 1,4;Q → 2,3,5;R → 1,4,6;S → 1,3,6

(C) P → 1,4,6;Q → 1,3,6;R → 2,3,5;S → 1,4

(D) P → 2,3,5;Q → 1,3,6;R → 1,3,6;S → 1,4,6

Space for Rough Work

SEC-3 : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

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1. The side of a cube is measured by vernier callipers (10 divisions of a vernier scale coincide with 9 divisions of main scale, where 1 division of main scale is 1 mm). The main scale reads 10 mm and first division of vernier scale coincides with the main scale. Mass of the cube is 2.736 g. Find the density (in g/cm^3) of the cube in appropriate significant figures.

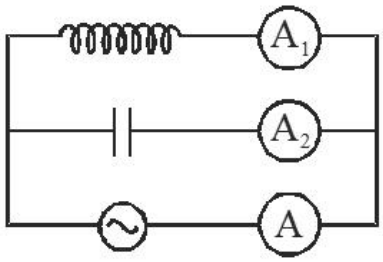
2. Three uniformly charged infinite wires with linear charge density λ are placed along x, y and z axis respectively. Find the flux of electric field through Gaussian surface given by $x^2 + y^2 + z^2 = 1$; $x > 0$; $y > 0$; $z > 0$. If your answer is $\frac{n\lambda}{12\epsilon_0}$ fill the value of n.

3. When a resistance R is connected in series with an element A, the electric current is found to be lagging behind the voltage by angle 30°. When the same resistance is connected in series with element B, current leads by 60°. When R, A, B are connected in series, the current now leads voltage by θ which is equal to $\tan^{-1}(K/\sqrt{3})$, then the value of K is (assume same AC source is used in all cases)

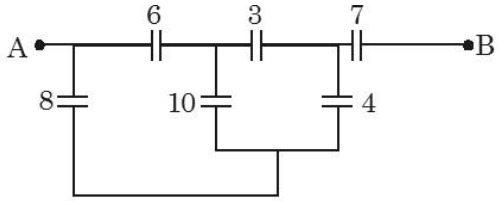
4. An x-ray beam of monochromatic photon are incident on a metallic surface having a negligible work function (take $\phi = 0$). It is seen that the wavelength of most energetic photoelectrons is equal to the wavelength of x-ray photons. Find the wavelength (in pm). Round off to nearest integer.

Space for Rough Work

5. Consider a circuit with an alternating source and contains inductor and capacitor. Given reading of A_1 and A_2 as 3 ampere and 5 ampere respectively. Find the reading of A in ampere.

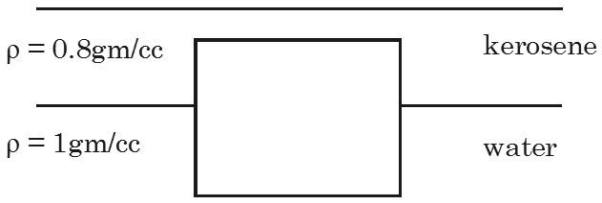


6. In the circuit diagram shown, all the capacitors are in μF. The equivalent capacitance between points A and B is (in μ F).



7. A sample of β active nuclei has a half life of 100 min. Initially, there are 10^{13} β active nuclei. Assume that all β particles emitted leave the sample. What is the charge acquired (in μC) by the sample in 200 min?

8. A cube is immersed in kerosene and water at rest without any external force as shown. To keep it completely submerged in water, we need to apply an extra force of 4 N. If it was initially submerged to depth of $\frac{a}{4}$ in kerosene, what is the length 'a' of the side of cube (in cm) ?



Space for Rough Work

CHEMISTRY

SEC-1 : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

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Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

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Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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1. Which statement(s) is/are correct -

(A) If 0.1M CH_3COOH aqueous solution ($K_a = 1.8 \times 10^{-5}$) is diluted at 25°C, then $[H^+]$ will increase.

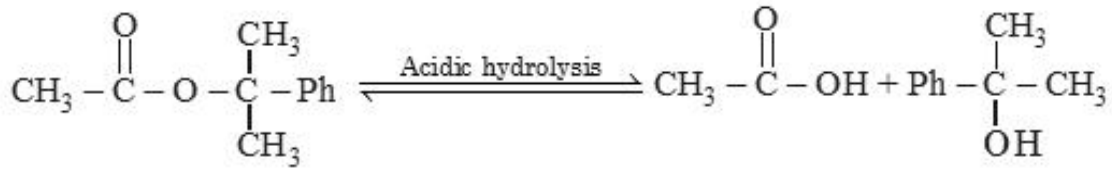
(B) On heating a sample of pure water $[H^+]$ and $[OH^-]$ increases but the sample still remains neutral

(C) 10^{-4}M HCl solution is less acidic than 0.1M HCN ($K_{a\ HCN} = 10^{-5}$) solution

(D) 10^{-8} M HCl aqueous solution will have pH exactly equal to 8

Space for Rough Work

2.



correct statements regarding the above reaction is/are :

(A) Free carbocation is involved in the reaction as an intermediate

(B) Alkyl-oxygen bond cleavage takes place

(C) Acyl-oxygen bond cleavage takes place

(D) Alcohol formed in the reaction will give immediate turbidity with Lucas reagent

3. Which of the following reaction is/are feasible?

(A) $2[Fe(CN)_6]^{3-} + 2OH^- + H_2O_2 \rightarrow 2[Fe(CN)_6]^{4-} + 2H_2O + O_2$

(B) $2K_4[Fe(CN)_6] + H_2O_2 + H_2SO_4 \rightarrow 2K_3[Fe(CN)_6] + K_2SO_4 + 2H_2O$

(C) $2KMnO_4 + 3H_2O_2 \rightarrow 2KOH + 2MnO_2 + 2H_2O + 3O_2$

(D) $2KMnO_4 + 5H_2O_2 + 3H_2SO_4 \rightarrow 2MnSO_4 + K_2SO_4 + 5O_2 + 8H_2O$

Space for Rough Work

4. Select the correct statement(s)?

(A) Addition of HgI_2 to aqueous solution of KI increases the freezing point since a complex $K_2[HgI_4]$ is formed

(B) Three aqueous solutions of ammonia labelled as A, B and C with concentrations 0.1 M, 0.01 M and 0.001 M respectively, then value of Van't Hoff factor for these solution will be in order :- $i_A < i_B < i_C$ (consider degree of dilution)

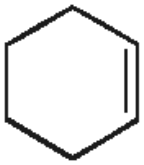
(C) Vapour density of $PCl_5(g)$ dissociating into PCl_3 (g) and $Cl_2(g)$ is 100. % dissociation of PCl_5 is 4.25% [Given : atomic mass of P = 31, Cl = 35.5]

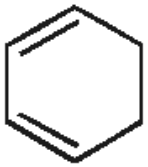
(D) If pH of 0.1 M monobasic acid is found to be 2, its osmotic pressure at a given temperature T K is 0.11 RT

5.

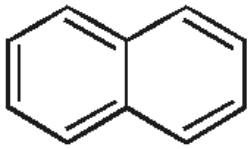
Compounds having heat of hydrogenation lesser than that of cyclohexa-1, 4-diene i.e. 

[Hint resonance energy of benzene is 36 Kcal]

(A) 

(B) 

(C) 

(D) 

Space for Rough Work

6. Which of the following compound/s does not liberate acidic gas during its hydrolysis.

(A) Ca_3P_2 (B) AlN (C) Al_2S_3 (D) CaH_2

Space for Rough Work

SEC-2 : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks : –1 In all other cases

1.

	List-I (Electrolysis using inert electrode)		List-II (Product at anode or cathode.)
P	Aqueous solution of sodium acetate	1	H_2
Q	Aqueous solution of sodium formate	2	CO_2
R	Concentrated aqueous solution of NaCl	3	Cl_2
S	Dilute aqueous solution of HCl	4	C_2H_6
		5	O_2

The correct option is -

(A) P → 1,2,4;Q → 1,2;R → 1, 3;S → 1, 5

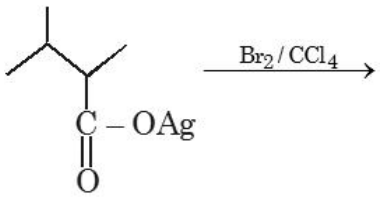
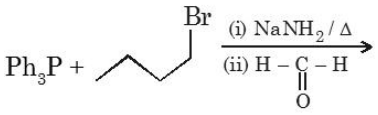
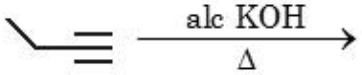
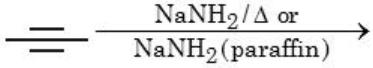
(B) P → 1,5;Q → 2,4;R → 1;S → 3, 4

(C) P → 1,4;Q → 2,4;R → 1, 5;S → 3, 5

(D) P → 2,4;Q → 1,4;R → 2, 5;S → 1,5

Space for Rough Work

2. Match the list

	List-I		List-II
P		1	Ionic organometalic product is formed
Q		2	Product can show electrophillic addition reaction with HBr (1 eq.)
R		3	Reaction involve free radical as an intermediate
S		4	Number of carbon in parent chain of product is 4

Select correct code for your answer :

(A) P → 3, 4;Q → 2;R → 2, 4;S → 1, 4

(B) P → 3, 4;Q → 2;R → 2, 4;S → 1, 3

(C) P → 3, 4;Q → 1, 2;R → 2, 4;S → 1, 4

(D) P → 2, 3;Q → 2;R → 2, 4;S → 1, 4

Space for Rough Work

3. Match the list:

	List-I (complex)		List-II (Properties)
P	$[CoF_6]^{3-}$	1	Diamagnetic and low spin complex
Q	$[Cr(C_2O_4)_3]^{3-}$	2	Paramagnetic and outer orbital complex
R	$[AuCl_4]^-$	3	Paramagnetic and inner orbital complex
S	$[Fe(CN)_6]^{3-}$	4	Paramagnetic and low spin complex

The correct option is

(A) P → 1;Q → 2;R → 3;S → 4

(B) P → 2;Q → 3;R → 1;S → 4

(C) P → 2;Q → 1;R → 3;S → 4

(D) P → 1;Q → 3;R → 2;S → 4

Space for Rough Work

4. Match the list:

	List-I (Complex Compound)		List- II (Properties)
P	$[Ni(H_2O)_6]Cl_2$	1	d^2sp^3 hybridisation
Q	$[Co(CN)_2(NH_3)_4]OC_2H_5$	2	Ionisation isomerism
R	$[IrCl_6]^{3-}$	3	μ = 2.83 B.M.
S	$[PtCl_2(NH_3)_4]Br_2$	4	$\Delta_0 < P$
		5	sp^3d^2 hybridisation

The correct option is

(A) P → 3,4,5;Q → 1,2;R → 1;S → 1,2

(B) P → 3,4;Q → 1;R → 2,3;S → 1,2

(C) P → 1,2,3;Q → 1,4;R → 4,5;S → 2,5

(D) P → 2,3;Q → 2;R → 3,4;S → 1

Space for Rough Work

SEC-3 : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

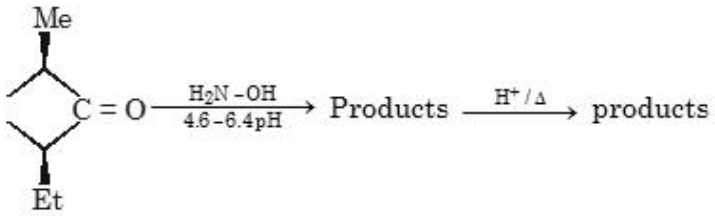
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1. For the first order reaction $A(g) \rightarrow 2B(g) + C(g)$ the half life for decomposition of A is 3 minute at 300K. Calculate the time (in minute) in which partial pressure of A(g) will drop from 2 bar to 0.5 bar at 400K. Given activation energy of reaction is 840 R. (Take ln2 = 0.7)

2. 2 moles of methane gas are compressed from (1 bar, 2L) to 2 bar isothermally against constant external pressure. Calculate work done in Joules.

3. For 10 minutes each, at 0°C, from two identical holes nitrogen and an unknown gas are leaked into a common vessel of 4 litre capacity. The resulting pressure in vessel is 2.8 atm and the mixture contains 0.4 mole of nitrogen. What is the molar mass of unknown gas?
(Take R = 0.0821 L –atm - mol^{-1} K^{-1})

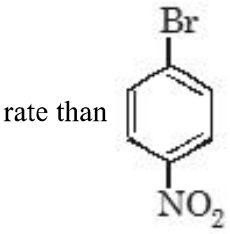
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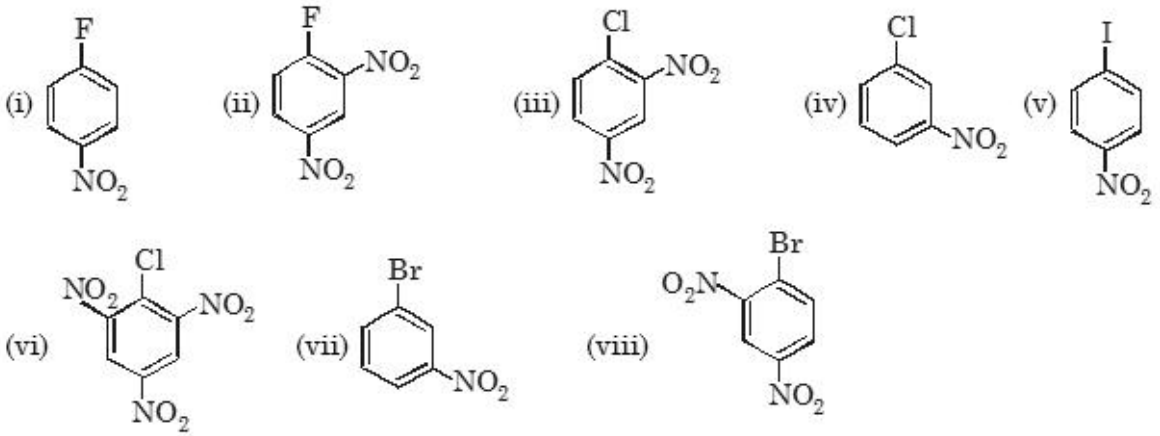
Find out total number of final products?

Space for Rough Work

5. How many of the following compounds given below have faster aromatic nucleophilic substitution

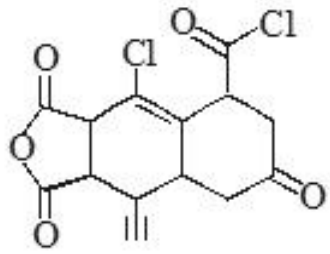


rate than



Space for Rough Work

6. Find out total number of moles of Grignard reagent RMgX consumed when treated it with 1 mole of 'X'



7. Find the number of electrons having $|m| \leq 1$, $l = 2$, $m_s = -\frac{1}{2}$ in Pd ($Z = 46$).

8. During extraction of iron, CO is the chief reducing agent. If 72 Kg of carbon is available and produced CO is completely used to reduce Haematite (present in excess amount) then calculate how much Kg of Iron is obtained after reduction.
(Given : atomic weight of C–12 g/mol, O–16 g/mol, Fe–55.85 g/mol)

Space for Rough Work

MATHEMATICS

SEC-1 : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

---

1. If e_1 is eccentricity of $9x^2 + 4y^2 = 36$ and e_2 is eccentricity of $9x^2 - 4y^2 = 36$ then

(A) $e_1^2 + e_2^2 < 3$ (B) $e_1^2 + e_2^2 > 3$ (C) $e_1^2 + e_2^2 < 4$ (D) $e_1^2 + e_2^2 > 4$

Space for Rough Work

2. Let $f : [0, 1] \to [0, \infty)$ be a differentiable function so that $f'(x)$ is decreasing function ; $f(0) = 0$, $f'(1) > 0$, $f'(0) > 0$ and $I = \int_0^1 \frac{dx}{1 + f^2(x)}\, dx$ then which of the following is **INCORRECT** ?

(A) $I \leqslant \frac{f(1)}{f'(1)}$ (B) $I < \frac{f(1)}{f'(1)}$ (C) $I \geqslant \frac{f(1)}{f'(1)}$ (D) $I > \frac{f(1)}{f'(1)}$

3. The coordinates of four angular points of a tetrahedron OABC are (0,0,0), (0,0,2), (0,4,0) and (6,0,0) respectively. A point P inside the tetrahedron is at the same distance r from the four plane faces of tetrahedron. Which of following **cannot** be the value of r ?

(A) $\frac{2}{3}$ (B) $\frac{2}{5}$ (C) 2 (D) 3

4. Let $C_k = {}^nC_k$ for $0 \le k \le n$ and $A_k = \begin{bmatrix} C_{k-1} & 0 \\ 0 & C_k \end{bmatrix}$ for $k \ge 1$ and

$A_1A_2 + A_2A_3 + A_3A_4 + ... + A_{n-1}A_n = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ then

(A) $a = b$ (B) $a + b = 2\,({}^{2n}C_{n-1} - n)$

(C) $a = 2b$ (D) $a + b = 0$

5. Probability that the length of a randomly chosen chord of a circle lies between $\frac{1}{2}$ and $\frac{3}{4}$ of its diameter is $\frac{p}{q}$ where p and q are coprime number, then

(A) $p + q = 21$ (B) $q - p = 11$ (C) $p + q = 20$ (D) $q - p = 10$

Space for Rough Work

6. For two complex numbers z_1 and z_2 ; $(az_1 + b\bar{z}_1)(cz_2 + d\bar{z}_2) = (cz_1 + d\bar{z}_1)(az_2 + b\bar{z}_2)$ if

(A) $\frac{a}{b} = \frac{c}{d}$

(B) $\frac{a}{d} = \frac{b}{c}$

(C) $|z_1| = |z_2|$

(D) $\arg(z_1) = \arg(z_2)$

Space for Rough Work

SEC-2 : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks : –1 In all other cases

---

1. Let $\phi(x) = 3f\left(\frac{x^2}{3}\right) + f\left(3 - x^2\right)\ \forall\ x \in (-3, 4)$ where $f''(x) > 0\ \forall\ x \in (-3, 4)$,

$g(x) = (a^2 - 3a + 2)\left(\cos^2\frac{x}{4} - \sin^2\frac{x}{4}\right) + (a - 1)x + \sin 1$

Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	$\phi(x)$ is increasing in	(1)	$(0, 1) \cup (1, 4)$
(Q)	$\phi(x)$ is decreasing in	(2)	$\left[-\frac{3}{2}, 0\right]\ ;\ \left[\frac{3}{2},\ 4\right)$
(R)	The set of values of a for which g(x) does not possess critical point is	(3)	$\left(-3,\ -\frac{3}{2}\right]\ ;\ \left[0, \frac{3}{2}\right]$
(S)	The set of values of a for which g(x) possess critical point is	(4)	$(-\infty, 0] \cup \{1\} \cup [4, \infty)$

(A) P → 2;Q → 3;R → 1;S → 4 (B) P → 2;Q → 3;R → 4;S → 1

(C) P → 3;Q → 2;R → 1;S → 4 (D) P → 3;Q → 2;R → 4;S → 1

Space for Rough Work

2. $S \equiv x^2 + y^2 - 2x - 4y + 1 = 0$ and $S' \equiv x^2 + y^2 - 4x - 2y + 1 = 0$ are equations of two circles.
Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	Sum of number of tangents from (1, 3) to S and number of tangents from (1,3) to S' is	(1)	1
(Q)	If r is radius of circle passes through point of intersection of S and S' and whose normal is the straight line $2x + y - 6 = 0$, then the value of $\frac{r^2}{4}$ is	(2)	2
(R)	If m is slope of common tangent of S = 0 and S' = 0 then $\|m\|$ is	(3)	3
(S)	If ℓ is length of common tangent of S and S' then the value of $2\ell^2$ is	(4)	4

(A) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$

(B) $P \rightarrow 2; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 4$

(C) $P \rightarrow 2; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$

(D) $P \rightarrow 1; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$

Space for Rough Work

3. (1) The sum of all the values of r satisfying ${}^{39}C_{3r-1} - {}^{39}C_{r^2} = {}^{39}C_{r^2-1} - {}^{39}C_{3r}$ is α_1.

(2) If ${}^{2n+3}C_{2n} - {}^{2n+2}C_{2n-1} = 15(2n+1)$ then the value of n is α_2

(3) If ${}^{56}P_{r+6} : {}^{54}P_{r+3} = 30800 : 1$ then the value of r is α_3

(4) ${}^{n+2}C_8 : {}^{n-2}P_4 = 57 : 16$ then the value of n is α_4

Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	The value of α_1 is	(1)	41
(Q)	The value of α_2 is	(2)	8
(R)	The value of α_3 is	(3)	14
(S)	The value of α_4 is	(4)	19

(A) P → 2;Q → 3;R → 1;S → 4

(B) P → 2;Q → 3;R → 4;S → 1

(C) P → 2;Q → 1;R → 3;S → 4

(D) P → 3;Q → 2;R → 1;S → 4

Space for Rough Work

4. (i) Let harmonic mean, arithmetic mean and geometric mean of two positive numbers a and b respectively be 4, A and G. If $2A + G^2 = 27$ then $a^2 + b^2 = \alpha_1$

(ii) If the value of $x + y + z = 15$ where a, x, y, z, b are in arithmetic progression while the value of $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ is $\frac{5}{8}$ where a, x, y, z, b are in harmonic progression, then the value of $a^2 + b^2$ is α_2

(iii) The value of $\sum_{r=1}^{16} \left(\frac{1^3 + 2^3 + \ldots\ldots + r^3}{1 + 3 + \ldots\ldots + (2r-1)} \right)$ is α_3

(iv) If the geometric mean and harmonic mean of two positive numbers x_1 and x_2 are 18 and $\frac{216}{13}$ respectively then the value of $|x_1 - x_2|$ is α_4.

Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	The value of α_1 is	(1)	52
(Q)	The value of α_2 is	(2)	45
(R)	The value of α_3 is	(3)	446
(S)	The value of α_4 is	(4)	15

(A) P → 2;Q → 1;R → 2;S → 3

(B) P → 2;Q → 2;R → 1;S → 4

(C) P → 2;Q → 1;R → 3;S → 4

(D) P → 1;Q → 1;R → 4;S → 2

Space for Rough Work

SEC-3 : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

---

1. In a triangle ABC (with usual notations), $a = 5$, $b = 4$ and $A - B = \cos^{-1}\left(\frac{31}{32}\right)$ then the value of $\frac{c}{5}$ is

2. In the expansion of $(1-x)^{30}(1+x+x^2)^{31}$ the coefficient of x^{47} is $(-{}^{30}C_r)$ then the value of $\left|\frac{r}{2}\right|$ is

3. In the equation $x^5 - 5x^4 + 9x^3 - 9x^2 + 5x - 1 = 0$ if p is the sum of real roots and q is the product of real roots then the value of $\frac{q}{p}$ is

4. Let $f : (0, \infty) \to (0, \infty)$ be a function differentiable at 3 and satisfying $f(3) = 3f'(3)$. If the value of $\lim\limits_{x\to\infty}\left(\frac{f\left(3+\frac{3}{x}\right)}{f(3)}\right)^x$ is L then $\frac{[L]}{5}$ is (where [.] is greatest integer function)

5. The value of $\int\limits_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{\cos x}{1-\cos x}\,dx$ is equal to (use $\pi = \frac{22}{7}$)

Space for Rough Work

6. Let $f(x)=\begin{vmatrix} \sec x & \cos x & \sec^2 x+\cot x \operatorname{cosec} x+\cos x \\ \cos^2 x & \cos^2 x & \operatorname{cosec}^2 x+\cos^4 x \\ 1 & \cos^2 x & \cos^2 x \end{vmatrix}$ then the value $\left|\int_0^{\pi/2} f(x)\,dx\right|$ is (use $\pi=\frac{22}{7}$)

7. If $y = 2x - 3$ is a tangent to the parabola $y^2 = 4a\left(x-\frac{1}{3}\right)$ then the value of $|a|$ is

8. OA, OB, OC are the sides of a rectangular parallelopiped whose diagonals are OO', AA', BB' and CC'. D is the centre of rectangle AC'O'B' and D' is centre of rectangle O'A'CB'. If sides OA, OB, OC are in ratio 1 : 2 : 3 and $\angle DOD' = \alpha$ then $\frac{697\cos^2\alpha}{128}$ is

(Centre of rectangle is mid point of diagonal)

Space for Rough Work

Space for Rough Work

English

(1001CJA102120083)

Test Pattern

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
FULL SYLLABUS
13-05-2021

JEE(Main+Advanced) : ENTHUSIAST COURSE [SCORE-II]

Time : 3 Hours **Maximum Marks : 240**

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **27** pages and that all the **20** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble ◯ **COMPLETELY.**
12. The correct way of darkening a bubble is as : ●
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **g = 10 m/s²** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

PHYSICS

SEC-1 : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

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1. Pick the correct option/options.

(A) If the sum of all the forces acting on a particle is zero, then it will necessarily be in equilibrium.

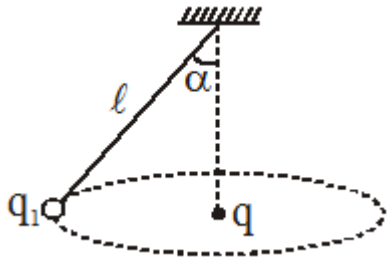
(B) When a sphere rolls on a horizontal fixed surface, there is no point on the sphere which has vertical velocity.

(C) When tall buildings are constructed on earth, the duration of day and night increases.

(D) A body is rotating non uniformly about a vertical axis fixed in an inertial reference frame. The resultant force on the particle of the body not on the axis is horizontal and skew with the axis.

Space for Rough Work

2. A charged ball of charge $+q_1$ is revolving around another charge $+q$ as shown in a conical pendulum. The motion is in a horizontal plane :-



(A) Tension in the string is greater than the weight of the ball.

(B) The tension in the string is greater than the electrostatic repulsive force.

(C) If the charge is removed, the speed of ball has to be increased to maintain the angle.

(D) If the charge is removed, the speed of ball has to be decreased to maintain the angle.

3. Mark the incorrect statement(s) for a standing wave established in a string :

(A) Energy of an element of medium remains constant.

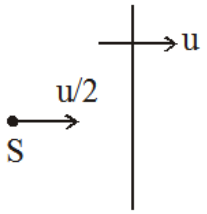
(B) Phase difference between any two particles can be $\frac{\pi}{2}$

(C) Energy can get transferred through antinodes.

(D) Energy can get transferred through nodes.

Space for Rough Work

4. A wall is moving with velocity u and a source of sound (frequency f) moves with velocity $\frac{u}{2}$ in the same direction as shown. Assuming that the sound travels with a speed of 10u, pick the right option or options.



(A) wavelength of incident sound is $\frac{19u}{2f}$.

(B) frequency of the incident sound is $\frac{18}{19}$f.

(C) wavelength of reflected sound is $\frac{u}{f}$.

(D) frequency of the reflected sound is $\frac{12f}{19}$.

5. A gas is undergoing an adiabatic process. At a certain stage, the values of volume and temperature are V_0 and T_0 respectively and the modulus of slope of V–T curve is m. Choose the correct options-

(A) The value of $C_V = \frac{RT_0m}{V_0}$

(B) The value of $C_P = \left(1 + \frac{T_0m}{V_0}\right) R$

(C) $m = \frac{V_0}{T_0(\gamma - 1)}$

(D) $m = \frac{-V_0}{T_0(\gamma - 1)}$

6. A long straight cylindrical shell has inner radius R_i and outer radius R_0. It carries current i, uniformly distributed over its cross-section. A solid wire of radius r is to be placed parallel to the cylinder axis, in the hollow region such that ($r < R_i$) :-

(A) The magnetic field is zero everywhere in hollow region. We conclude that the wire is on the cylinder axis and carries current i in the same direction as current in the shell.

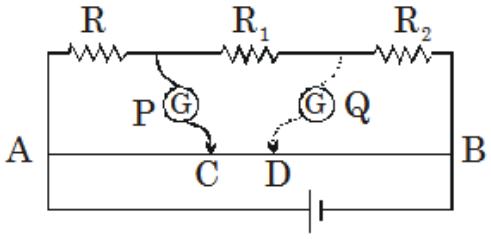
(B) The magnetic field is zero everywhere outside the shell ($r > R_0$). We conclude that the wire is on the cylinder axis and carries i in the direction opposite to the direction of current in the shell.

(C) The magnetic field is zero everywhere in the hollow region. We conclude that the wire may be anywhere in the hollow region but must be carrying current i in the same direction as the current in the shell.

(D) The magnetic field is zero everywhere in the hollow region. We conclude that wire does not carry any current.

Space for Rough Work

7. The diagram shows a modified meter bridge, which is used for measuring two unknown resistance at the same time. When only the first galvanometer P is used, for obtaining the balance point, it is found at point C. Now the first galvanometer P is removed and the second galvanometer Q is used, which gives balance point at D. Using the details given in the diagram, find out the value of R_1 and R_2.



AB=L; AC=L/4 and AD=2L/3

(A) $R_1 = 5R/3$ (B) $R_2 = 4R/3$ (C) $R_1 = 4R/3$ (D) $R_2 = 5R/3$

8. A mirror is hanging by a light string of length ℓ from ceiling. A light pulse of monochromatic light rays is incident normally on the mirror and gets reflected. Which of the following statements is/are correct ?

(A) Mirror will start to oscillate.

(B) Wavelength of reflected rays will be greater than that of the incident rays.

(C) Wavelength of reflected rays may be less than that of incident rays.

(D) Assuming the intensity of incident light to be constant, the maximum angle to which the mirror deflects depends on the wavelength of the light pulse.

Space for Rough Work

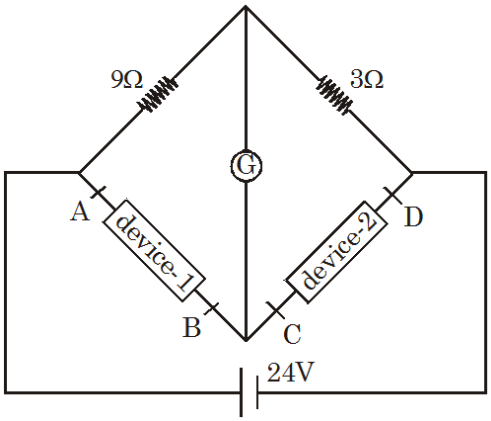
SEC-2 : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : –2 In all other cases.

Paragraph for Questions 1 and 2

For given circuit, give answer of following questions



1. If device–1 is battery of emf 10V and 2Ω internal resistance (+ive end at A) and device-2 is battery of emf x volt and internal resistance 1Ω (+ive end at C) and reading of galvanometer is zero then value of x is :-

(A) 6 volt (B) 4 volt (C) 10 volt (D) 2 volt

2. If device-1 and device-2 are capacitors of capacitance C_1 and C_2 and reading of galvanometer remain zero then $\frac{C_1}{C_2}$ is (no charge pass through Galvanometer):-

(A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) 1 (D) $\frac{2}{3}$

Space for Rough Work

Paragraph for Questions 3 and 4

A charged particle having mass 'm' and charge '+q' is released from rest at origin in uniform electric field $\vec{E} = E_0\hat{j}$ and uniform magnetic field $\vec{B} = B_0\hat{k}$.

3. Maximum value of 'y' coordinate of particle in the path.

(A) $\frac{mE_0}{qB_0^2}$ (B) $2\frac{mE_0}{qB_0^2}$ (C) $\frac{3mE_0}{qB_0^2}$ (D) $\frac{4mE_0}{qB_0^2}$

4. Mark the correct option(s) :

(A) When y-coordinate is maximum, the speed of the particle is maximum.

(B) When y-coordinate is maximum, the velocity of particle is in x-direction.

(C) Radius of curvature of path when y-coordinate is maximum is $\frac{4mE_0}{qB_0^2}$.

(D) Radius of curvature of path when y-coordinate is maximum is $\frac{2mE_0}{qB_0^2}$.

Space for Rough Work

SEC-3 : (Maximum Marks: 32)

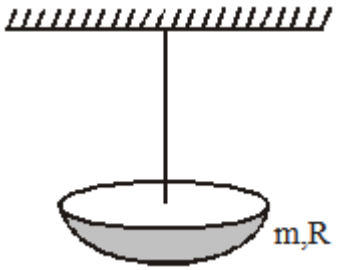
- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

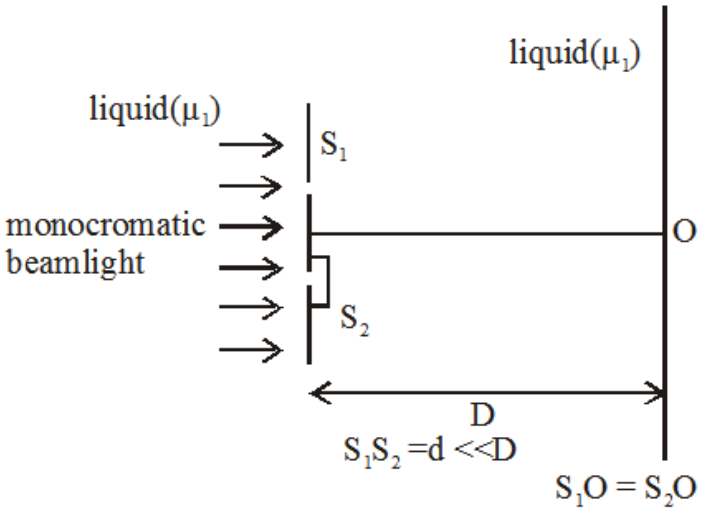
Negative Marks : –0 In all other cases

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1. A uniform solid hemisphere of mass m and radius R is attached to the roof with a chord of torsional constant C and performing torsional SHM. Then the time period (in seconds) of SHM is (Take m = 15 kg, R = $\frac{2}{\pi}$m, C = 6 Nm/rad).



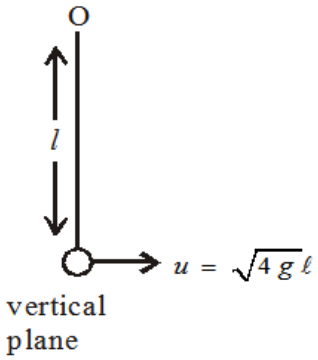
2. Young's double slit experiment is conducted in a liquid of refractive index μ_1 as shown in figure. A thin transparent slab of refractive index μ_2 is placed in front of the slit S_2. If $\mu_2 = \frac{3}{2}, \mu_1 = \frac{4}{3}$, t = 8 μm, the magnitude of optical path difference (wrt liquid) at 'O' in μm is :



Space for Rough Work

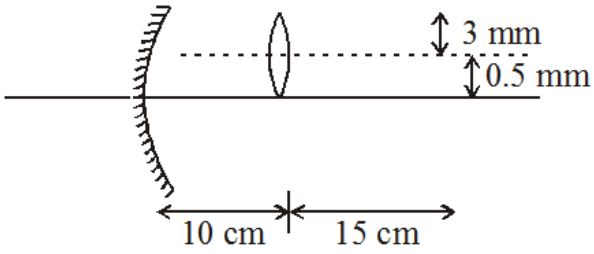
3. We would like to increase the length of a 15 cm long copper rod of cross-section 4 mm^2 by 1mm. The energy absorbed by the rod if it is heated is E_1. The energy absorbed by the rod if it is stretched slowly is E_2. The ratio of E_1 and E_2 is α. The value of $\frac{\alpha}{100}$ is. [Various parameters of Copper are: Density = 9×10^3 kg/m^3, Thermal coefficient of linear expansion = 16×10^{-6} K^{-1}, Young's modulus = 135×10^9 Pa, Specific heat = 400 J/kg-K]

4. A pendulum bob hinged at O is given velocity $u = \sqrt{4g\ell}$ from bottommost point as shown. If ℓ = 54 cm, find maximum height (in m) reached by bob from point of projection.



Space for Rough Work

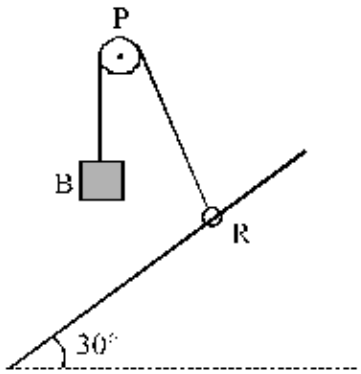
5. A concave mirror of focal length 20 cm and a convex lens of focal length 10 cm are kept with their optic axis parallel but separated by 0.5 mm as shown in figure. The distance between lens and mirror is 10 cm. An object of height 3 mm is placed on the optic axis of lens at distance 15 cm from the lens. Find height of image formed by mirror after refraction from lens (in mm).



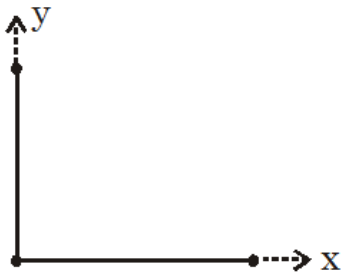
6. A uniform disc of radius R having charge Q distributed uniformly all over its surface is placed on a smooth horizontal surface. A magnetic field $B = Kxt^2$, where K = constant, x is the distance (in metre) from the centre of the disc and t is the time (in second) is switched on perpendicular to the plane of the disc. The torque (in N-m) acting on the disc after 15 sec. (Take 2KQ = 1 S.I. unit and R = 1 metre).

Space for Rough Work

7. A small ring R of mass 0.2 kg is threaded onto a fixed rod which is inclined at an angle of 30° to the horizontal. The ring is attached to a block B of mass 0.5 kg by means of a light string which passes over a smooth pulley P. The part RP of the string is perpendicular to the rod, and the part PB of the string is vertical (see diagram). The system is in equilibrium. Assuming the equilibrium is limiting, if the coefficient of friction (μ) between R and the rod is $\mu = \frac{1}{\alpha - \sqrt{\beta}}$ where α and β are the smallest possible integer. Fill the value of (α + β).



8. Two uniform rods of same length but different density are arranged as shown in the diagram. Centre of mass of this arrangement lies on a line ax + by = ℓ. (ℓ is the length of rod) find a + b = ?



Space for Rough Work

CHEMISTRY

SEC-1 : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

---

1. For 0.1 M aq. solution of Na_2CO_3 at 25°C, select the correct options
(Use : H_2CO_3 : $\kappa_{a_1} = 10^{-5}$, $\kappa_{a_2} = 10^{-9}$) and K_w (H_2O) = 10^{-14}

(A) pH of the solution is 11.

(B) Equilibrium concentration of H_2CO_3 is 10^{-9} M.

(C) Equilibrium concentration of HCO_3^- is 10^{-3} M

(D) Equilibrium concentration of CO_3^{2-} is 0.01 M

2. Select the incorrect options :

(A) Gold sol is a positive charged sol.

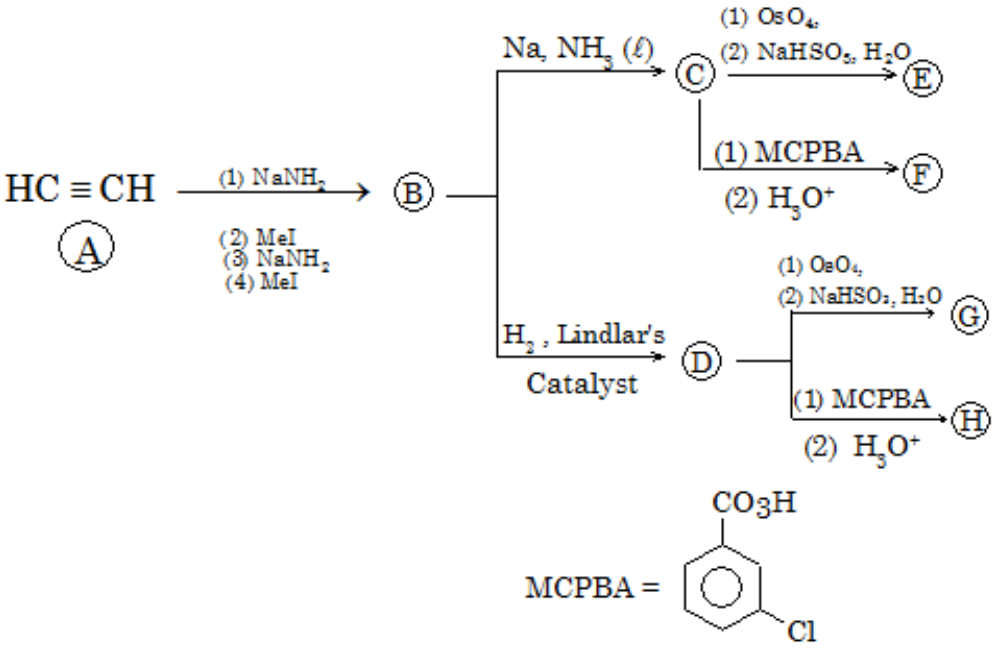
(B) Lyophobic sols can act as protective colloid for Lyophilic sols.

(C) When $FeCl_3$ is hydrolysed with hot water, then Ferric hydroxide sol (positively charged) will be obtained.

(D) Blood is negatively charged sol.

Space for Rough Work

3.



Select the correct statements(s) for above reaction sequence ?

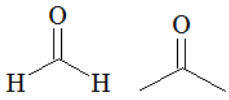
(A) F and G are identical.

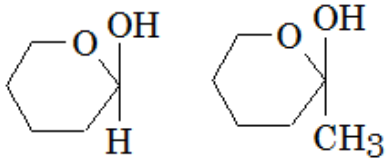
(B) E and H are optically inactive due to external compensation.

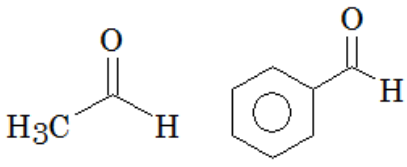
(C) Compound (A) and compound (B) can be differentiated by Tollen's reagent

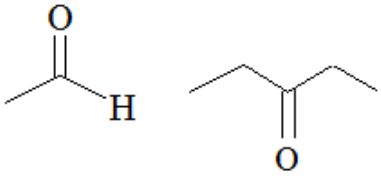
(D) F and G are enantiomers

4. Among the following pair of compounds, identify the pair(s) which can be differentiated by Fehling as well as Tollen's reagent ?

(A) 

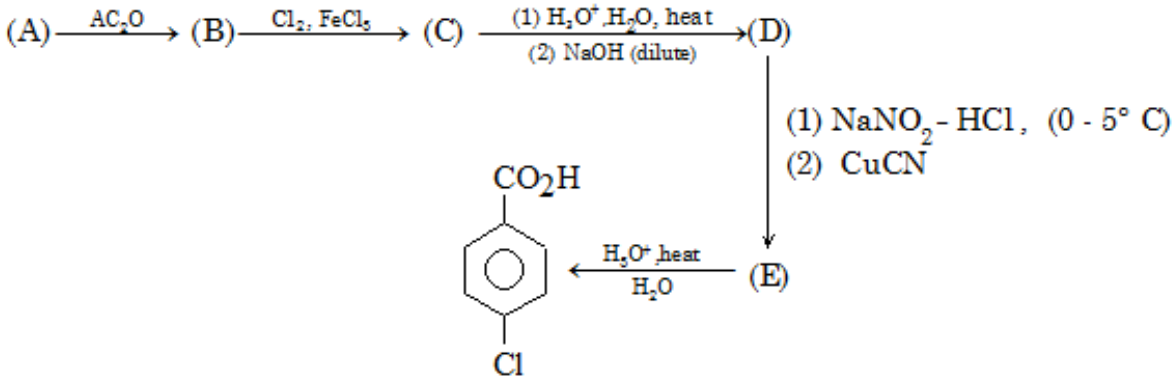
(B) 

(C) 

(D) 

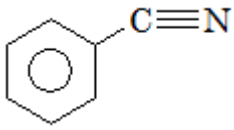
Space for Rough Work

5. Compound (A) has molecular formula C_6H_7N, undergoes following reaction sequence.

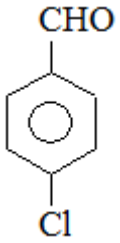


all are major organic product in above reaction sequence. Select the correct statement(s) for the reaction sequence.

(A) Order of basic strength is A > D > E .

(B) Compound (A) on reaction with $CHCl_3$, Alc. KOH results formation of 

(C) Reactivity order towards aromatic electrophilic substitution is A > B.

(D) Compound (E) on reaction with $SnCl_2$, HCl followed by H_3O^+ gives 

as a major product.

Space for Rough Work

6. Which of the following is/are true for the complex compound with IUPAC name as :
Dichloridobis(ethane-1, 2-diamine)cobalt(III) sulphate

(A) Its formula is $\left[Co\left(\begin{matrix}CH_2-NH_2\\|\\CH_2-NH_2\end{matrix}\right)_2 Cl_2\right] SO_4$

(B) The co-ordination number of central metal ion is 6

(C) The effective atomic number of central metal is 32

(D) The compound can be categorised as a heteroleptic complex

7. The compound(s) that give more than one gaseous product on strong heating with Bunsen burner is/are

(A) Baking soda (B) Washing soda

(C) Chile salt petre (D) Calcium oxalate

8. The reaction Fe + dilH_2SO_4 $\rightarrow$; gives the product(s)

(A) $FeSO_4$ (B) $Fe_2(SO_4)_3$ (C) H_2 (D) SO_2

Space for Rough Work

SEC-2 : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

Paragraph for Questions 1 and 2

For reversible Galvanic cell, thermodynamic parameters are given as :

$(\Delta G)_{T,P} = \Delta H - T\Delta S$

$(\Delta G)_{T,P} = -nFE_{cell}$

$\Delta S = nF\left[\frac{dE_{cell}}{dT}\right]_P$

$W_{usefull} = -(\Delta G)_{T,P}$

Answer the following questions :

1. For given cell Pt | H_2 | HCl | AgCl | Ag, variation of e.m.f. of the cell with temperature is given as

E_{cell} (in volt) = $0.4 - 2 \times 10^{-6}\left(\frac{T}{K}\right)^2$

(T = temperature in Kelvin) then ΔG for given cell reaction at 27°C will be

$\frac{1}{2}H_2 + AgCl \rightarrow Ag + H^+ + Cl^-$

(A) – 0.44 F (B) – 0.4 F (C) – 0.22 F (D) – 0.33 F

2. For above cell reaction, entropy change per mole of Ag at 27°C will be
(Use : 1F = 96500 C)

(A) – 115.8 J/K (B) – 231.6 J/K (C) – 57.6 J/K (D) – 463.2 J/K

Space for Rough Work

Paragraph for Questions 3 and 4

The hydrolysis reactions of halides can take place at room temperature or slightly higher temperature when the vacant orbitals is/are available on the central atom and some times on the bonded atoms.

3. Which compounds undergo partial hydrolysis :

(A) BF_3 (B) $SiCl_4$ (C) SiF_4 (D) $SOCl_2$

4. Which of the following can be hydrolysed at room temperature :

(A) PCl_3 (B) PCl_5 (C) NF_3 (D) NCl_3

Space for Rough Work

SEC-3 : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Negative Marks : –0 In all other cases

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1. A gaseous mixture of H_2 and He in the mass ratio 1 : 4 is successively effused through a series of containers connected through narrow tubes of negligible volume and fitted with valves. The valves are successively opened for a very small time interval, dt. Find the minimum number of total containers required to obtain a molar ratio 8 : 1 for H_2 and He after the effusion process.

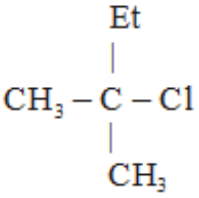
2. One mole of an diatomic ideal gas in an insulated cylinder fitted with a massless, frictionless movable piston is expanded gradually from 1 L to 32 L. If the initial temperature of gas is 1000 K then find magnitude of change in Gibb's Free energy in kcal.
(Molar entropy of gas = 3 cal mol^{-1}/k^{-1})
(Use, R = 2 cal mol^{-1} k^{-1})
(Assume molar entropy of a gas is constant)

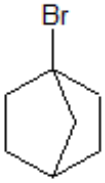
3. The activation energy of a reaction is given as $E_a\left(\text{cal mol}^{-1}\right) = 0.01T^2 + \frac{20}{7}T$ where T is temperature in kelvin. Find value of $\ell n\frac{k_2}{k_1}$ where k_1 and k_2 are rate constants at temperature 200 K and 400 K respectively. (Use 2 = 0.7 and R = 2 cal mol^{-1} k^{-1})

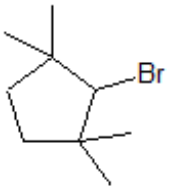
4. A polyhydroxy compound having molecular mass 150, on reacting the compound with CH_3COCl, the precipitate obtained has molecular mass of 318. Calculate number of OH group present in compound :

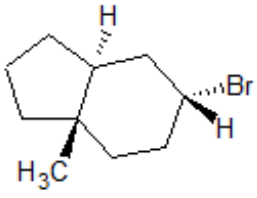
Space for Rough Work

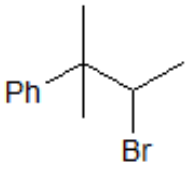
5. Examine the given structure and select those that satisfy given conditions

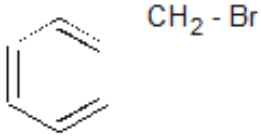
(a) 

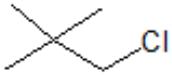
(b) 

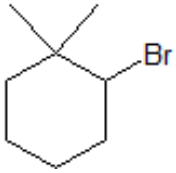
(c) 

(d) 

(e) 

(f) 

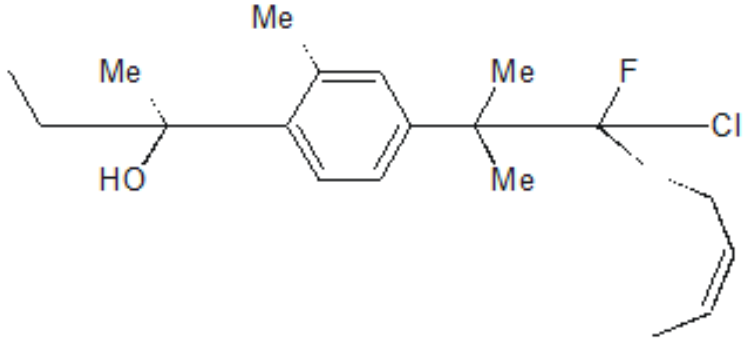
(g) 

(h) 

Find out no. of compounds which give SN^2 reaction with in $CH_3\overset{\ominus}{S}\,\overset{\oplus}{Na}$ in DMF faster with respect to CH_3CH_2Br

Space for Rough Work

6. (a) Total number of stereoisomers of given compound are x



(b) Number of optically active stereoisomers of 1,2,3,4,5,6-hexachlorocyclohexane is y.

Then x – y is

7. In the list of compounds given below, identify the number of compounds that have $d\pi - p\pi$ bonding in the molecular structure with sp^2 hybridization of the underlined atom.

$\underline{C}Cl_4$, $\underline{P}OCl_3$, $\underline{S}O_2$, $\underline{S}O_3$, $H_2\underline{S}O_4$, $\underline{S}OCl_2$, $\underline{S}O_2Cl_2$, $\underline{B}F_3$, $\underline{B}Cl_3$, $\underline{N}(SiH_3)_3$

8. How many of the following is/are yellow and water insoluble substances ?

Ag_2CrO_4,	$PbCrO_4$,	Na_2CrO_4,	$BaCrO_4$,	CrO_2Cl_2,
$(NH_4)_2Cr_2O_7$,	$(NH_4)_3PO_4.12MoO_3$,	$K_3[Co(NO_2)_6]$,	AgI,	PbI_2

Space for Rough Work

MATHEMATICS

SEC-1 : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

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1. If $f(x) = 1 - (\sin 2x)\sqrt[4]{2\sqrt{\tan^2 x((\tan^2 x)+6)+9} + \tan^4 x - 5}$, where $x \in \left(\frac{3\pi}{2}, 2\pi\right]$

(A) f(x) = 1–2 sin x

(B) f(x) = 1 + 2 cos x

(C) f(x) can take only 2 integral value

(D) f(x) can take only 3 integral value

2. x + y = 2 and x – y = 2 are tangents on a parabola at (1, 1) and (4, 2) respectively. Which of the following is/are correct

(A) Equation of directrix is x + 3y = 2

(B) Equation of axis is 3x – y = 5

(C) Focus of the parabola is at $\left(\frac{8}{5}, \frac{6}{5}\right)$

(D) Vertex of the parabola is at $\left(\frac{33}{20}, \frac{13}{20}\right)$

Space for Rough Work

3. Probability of n heads in 2n tosses of a fair coin can be given by

(A) $\prod_{r=1}^{n}\left(\frac{2r-1}{2r}\right)$ (B) $\prod_{r=1}^{n}\left(\frac{n+r}{2r}\right)$ (C) $\sum_{r=0}^{n}\left(\frac{{}^{n}C_r}{2^n}\right)^2$ (D) $\frac{\sum_{r=0}^{n}({}^{n}C_r)^2}{\left(\sum_{r=0}^{2n}{}^{2n}C_r\right)}$

4. P(n) denotes the number of natural number less than n, which are relatively prime with n, then

(A) $P\left(2^2.5\right)=2^3$ (B) $P\left(2^3\right)=2^2$

(C) $P\left(2^2.3\right)=2^2$ (D) $P\left(2.3^2\right)=2^2.3$

5. If the line y = 2 – x is tangent to the circle 's' at a point P(1, 1) and circle 's' is orthogonal to the circle $x^2+y^2+2x+2y-2=0$ then which of the following statement is true about circle 's'.

(A) The equation of the circle 's' is $x^2+y^2-x-y=0$

(B) The length of the tangent from (2, 2) to the circle 's' is '2'

(C) The equation of the director circle of circle 's' is $x^2+y^2-2x-2y+2=0$

(D) The length of the common chord of given circle and circle 's' will be $\frac{\sqrt{10}}{3}$

Space for Rough Work

6. If $I_m = \int_0^{m\pi} x\,|\sin x|\, e^{\cos 4x}\,dx$ and $J_n = \int_0^{n\pi} |\cos x|\, e^{\cos 4x}\,dx$, where $m, n \in I$ then

(Note : [k] denote GIF)

(A) $\left[\frac{I_1}{J_1}\right] = 0$ (B) $\left[\frac{I_2}{J_2}\right] = 3$ (C) $\left[\frac{I_2}{J_1}\right] = 0$ (D) $\left[\frac{I_3}{J_3}\right] = 4$

7. Let $f(x) = \cos x \sin 2x$, then

(A) $\min f(x) > -\frac{7}{9} \forall x \in (-\pi, \pi)$ (B) $\min f(x) > -\frac{9}{7} \forall x \in (-\pi, \pi)$

(C) $\min f(x) > -\frac{1}{9} \forall x \in [-\pi, \pi]$ (D) $\min f(x) > -\frac{2}{9} \forall x \in [-\pi, \pi]$

8. Area bounded by $\sqrt{x+y+1} < \sqrt{y-x+2}$ & $y < 0$ is less than

(A) $\frac{9}{16}$ (B) $\frac{9}{5}$ (C) $\frac{5}{4}$ (D) $\frac{16}{5}$

Space for Rough Work

SEC-2 : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

Paragraph for Questions 1 and 2

A square matrix $A = [a_{ij}]$ is called symmetric matrix if $a_{ij} = a_{ji} \ \forall \ i, j$ and a square matrix $A = [a_{ij}]$ is called skew -symmetric matrix if $a_{ij} = -a_{ji} \ \forall \ i, j$.

1. If A and B are symmetric matrices of order 3 then, Which of the following statements are ALWAYS TRUE.

I) A^{2n+1} is a skew-symmetric matrix for $n \in N$

II) AB + BA is a symmetric matrix,

III) AB – BA is a skew-symmetric matrix

IV) Det(B) is non zero.

(A) I only

(B) II & III

(C) I, II & III

(D) I, II , III & IV

2. $A = \begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 5 \\ 3 & -5 & 0 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Then AX = B, will have

(A) Exactly one solution for X

(B) No solution for X

(C) Multiple solutions for X

(D) None of these

Space for Rough Work

Paragraph for Questions 3 and 4

Linked comprehensive type passage:

Equation of line L = 0 is given by $2x - y + z - 2 = 0 = x + 2y - z - 3$, then

3. The equation of the plane containing the line L = 0 and passing through the point $(3, 2, 1)$ is ax + by + cz + 1 = 0, then a – b + c is divisible by

(A) 3 (B) 4 (C) 2 (D) 6

4. Vector equation of the plane which passes through the point $(-1, 3, 2)$ and is perpendicular to line L = 0 is

(A) $\vec{r}.(\hat{i} - 3\hat{j} - 5\hat{k}) + 20 = 0$ (B) $\vec{r}.(\hat{i} - 3\hat{j} + 5\hat{k}) = 0$

(C) $\vec{r}.(\hat{i} + 3\hat{j} + 5\hat{k}) - 18 = 0$ (D) $\vec{r}.(-\hat{i} + 3\hat{j} - 5\hat{k}) + 3 = 0$

Space for Rough Work

SEC-3 : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Negative Marks : –0 In all other cases

---

1. A tangent is drawn to the curve, $\frac{x^2}{16} + \frac{y^2}{9} = 1$ at the point P meeting the co-ordinate axis in T and t. If perpendicular from origin O to the about tangent meet at Y, then value of the product (Tt) (PY) is equal to

2. A continuous function $f : \left[0, \frac{3}{4}\right] \to R$ satisfies the equation $f(x) = (1 + x^2)\left(1 + \int_0^x \frac{f^2(t)}{1+t^2}dt\right)$, then $\left[f\left(\frac{1}{2}\right)\right]$ is

[Note : [K] denotes greatest integer less than or equal to K.]

3. The quadratic polynomial $P(x)$ satisfy the equation $P(x) - P(x + 1) = 2x - 1 \ \forall \ x \in R$. If the greatest value of $P(x)$ in $(-\infty, 0]$ is 2, then value of $P(-1) + P(1)$ is equal to

---

Space for Rough Work

4. $S_n = \lim_{x\to\infty}\left(\left(\left(x+\frac{1}{\sqrt{4n^2}}\right)\left(x+\frac{1}{\sqrt{4n^2-1^\alpha}}\right)\left(x+\frac{1}{\sqrt{4n^2-2^\alpha}}\right)....\left(x+\frac{1}{\sqrt{4n^2-(n-1)^\alpha}}\right)\right)^{1/n}-x\right)$

,$n\in N$

and for $\alpha = 1$, $L = \lim_{n\to\infty}(n.S_n)$ and

for $\alpha = 2$, $M = \lim_{n\to\infty}(n.S_n)$

then value of $[L + M]$ is equal to

[Note : [y] denotes the greatest integer function of y.]

5. Let a real valued function $f(x) = x^4 + 4x^3 - 8x^2 + P$, $P \in R$. If λ is the least value of P for which $g(x) = |f(x)|$ is differentiable $\forall\, x \in R$, then value of $\frac{\lambda}{16}$ is equal to

6. Let A and B be two sets such that n(A) = 4 and n(B) = 3. If (x, 1), (y, 2), (z, 1) and (2, 3) are in A × B, where x, y, z are distinct elements. If a function f : A → B defined & number of onto function from set A to set B is λ, then value of $\frac{\lambda}{6}$ is equal to

7. Let 3 vector $\vec{\ell}, \vec{m}$ & $\vec{n}$ are such that $|\vec{\ell}| = 3, |\vec{m}| = \frac{1}{2}$ & $\vec{\ell}.\vec{m} = \vec{m}.\vec{n} = \vec{n}.\vec{\ell} = 0$.

If $\left[\vec{\ell}\times\left(\vec{m}\times\vec{\ell}\right)\quad \vec{n}\times\vec{m}\quad \vec{n}\right] = 36$, then value of $|\vec{n}|$ is equal to

8. Let a circle with centre O and radius k. 3 points P, Q, R are chosen on a circle such that if length of chord PQ, QR, RP are x, y & z respectively, $x^2 + y^2 + z^2 = 5$, $\angle QOR = 60^\circ$ and Area of ΔPQR is $\frac{\sqrt{3}}{4}$ unit, then the value of radius of circle k will be equal to

Space for Rough Work

Space for Rough Work

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2020 - 2021)

JEE (Advanced)
MAJOR
13-05-2021

JEE(Main + Advance) : IIT-JEE(M + A) ENTHUSIAST & LEADER ENTHUSIAST & LEADER (PHASE : ALL PHASES)

Test Type : MAJOR Test Pattern : JEE (Advanced)

TEST DATE : 13-05-2021

ANSWER KEY PAPER-1

PHYSICS

SEC-1	Q.	1	2	3	4	5	6		
	A.	B,D	C	B,C	A,C,D	A,B,C,D	A,C		
SEC-2	Q.	1	2	3	4				
	A.	A	B	C	D				
SEC-3	Q.	1	2	3	4	5	6	7	8
	A.	2.66	9.00	2.00	1.00	2.00	2.80	1.20	20.00

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6		
	A.	B,C	A,B,D	A,B,C,D	A,B,C,D	A,B,C	A,B,D		
SEC-2	Q.	1	2	3	4				
	A.	A	A	B	A				
SEC-3	Q.	1	2	3	4	5	6	7	8
	A.	3.00	200.00	448.00	2.00	5.00	6.00	6.00	223.40

MATHEMATICS

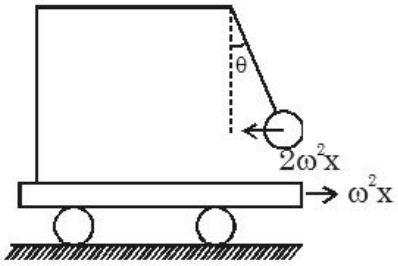
SEC-1	Q.	1	2	3	4	5	6		
	A.	B,C	C,D	B,C,D	A,B	A,B	A,D		
SEC-2	Q.	1	2	3	4				
	A.	A	B	A	C				
SEC-3	Q.	1	2	3	4	5	6	7	8
	A.	1.20	7.50	0.25	0.40	0.42, 0.43	1.45	4.66, 4.67	4.50

HINT – SHEET

PHYSICS

SEC-1

2. Ans (C)



$1.8 \times \frac{\pi}{180} \times \ell \times x = 3A$

$A = \frac{\pi\ell}{150} = \frac{\pi}{150} \times \frac{3}{10} = \frac{\pi}{500}m$

Centre of mass is at rest.

$3 \times x = 2l\theta$

$x = \frac{2\ell\theta}{3}$

from carts frame

$-(2g\,\ell\sin\theta + 2 \times \omega^2 x \times \ell) = 2\ell^2\alpha$

$-(2g\ell\theta + 2 \times \omega^2\ell \times 2\frac{\ell\theta}{3}) = -2\ell^2 \times \omega^2\theta$

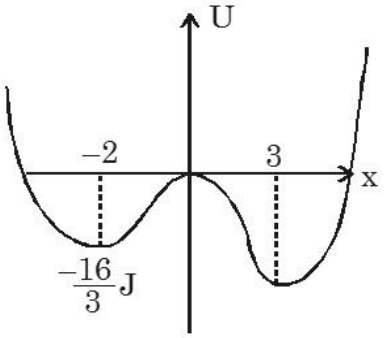
$+ 2g\ell + \frac{4}{3}\omega^2\ell^2 = 2\omega^2\ell^2$

$2g\ell = \frac{2}{3}\omega^2\ell^2$

$\omega = \sqrt{\frac{3g}{\ell}} \Rightarrow T = 2\pi\sqrt{\frac{\ell}{3g}} =$

$2\pi\sqrt{\frac{0.3}{300}} = \frac{\pi}{5}\text{sec}$

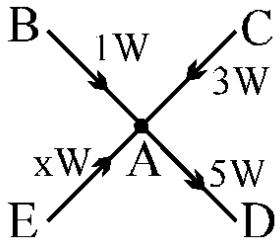
3. Ans (B,C)



To reach x = 3, it has to cross x = 0. Energy required = $\frac{16}{3}$J

4. Ans (A,C,D)

Heat flow



At A

$x + 1 + 3 = 5$

$x = 1$

heat in flows from E

$\therefore T_E > T_A$

$\therefore T_C > T_A > T_D$

$T_B - T_A = T_E - T_A$

$\therefore T_B = T_E$

5. Ans (A,B,C,D)

$R = v_x t$

$t = \sqrt{\frac{2(1-y)}{g}}$

$v_x = \sqrt{2gy}$

$\Rightarrow 0.8 = 2\sqrt{y(1-y)}$

$0.4^2 = y(1-y)$

$y^2 - y + 0.16 = 0$

$y = \frac{1 \pm \sqrt{1-0.64}}{2} = \frac{1 \pm 0.6}{2}$ = 0.8 m or 0.2 m

6. Ans (A,C)

$I_0 = 4I_0\cos^2\frac{\pi yd}{\lambda D}$

$\cos\frac{\pi yd}{\lambda D} = \pm\frac{1}{2}$

$\frac{\pi yd}{\lambda D} = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}.....$

$y = \frac{\beta}{3}, \frac{2\beta}{3}, \frac{4\beta}{3}$ .....

$\Delta y = \frac{\beta}{3}$ or β or $\frac{2\beta}{3}$ or $\frac{4\beta}{3}$

$\Delta y = (\mu - 1)\frac{tD}{d} = \frac{\pi}{3}\frac{\lambda D}{d}$

$\Rightarrow t = \frac{2n\lambda}{3} = 300\text{ nm} \times n$

PHYSICS

SEC-2

1. Ans (A)

Fundamental idea of electric & magnetic forces on charged particles.

$\vec{F}_e = q\vec{E}$

$\vec{F}_m = q(\vec{V} \times \vec{B})$

2. Ans (B)

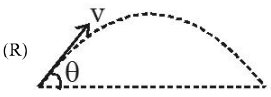
$v > 0$ and $a > 0$
$v < 0$ and $a < 0$ $\Big\}$ → speeding up ↑

$v > 0$ and $a < 0$
$v < 0$ and $a > 0$ $\Big\}$ → speeding down ↓

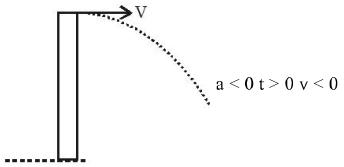
(P) $v = \frac{dx}{dt}$, $a = \frac{dv}{dt}$ at $t = 0 \Rightarrow v > 0$ and $a < 0$ (Q) $x = t^2 + 3t$

$v = \frac{dx}{dt} = 2t + 3$ at $t = 0 \Rightarrow v = 3$m/s $a = \frac{dv}{dt} = 2$ (remain constant)

at $t = 0 \Rightarrow v > 0$ and $a > 0$

(R) 

$a < 0$ at $t = 0 v > 0$ (S)



3. Ans (C)

(A) B → C is isobaric expansion

$T_B < T_C$ and $V_B < V_C$

So $\Delta U > 0$ and $\Delta W > 0$; Hence $\Delta Q > 0$

(B) A → B is isochoric heating

$T_B > T_C$ and $V_B = V_A$

So $\Delta U > 0$ and $\Delta W = 0$; Hence $\Delta Q > 0$

(C) A → B → C → A is cyclic process

$\Delta T = 0$ So, $\Delta U = 0$ But $\Delta W > 0$ and Hence $\Delta Q > 0$

(D) C → A

$T_A < T_C$ and $V_A < V_C$

So $\Delta U < 0$ and $\Delta W < 0$, So $\Delta Q < 0$

4. Ans (D)

F = +20cm u = −30cm V = $\frac{uF}{u+F} = \frac{(-30)(20)}{-30+20} = \frac{+600}{-10} = +60$cm m $= \frac{v}{u} = \frac{60}{-30} = -2$

Real, inverted and magnified (Q) $F = -40$ cm

$u = -30$ cm $v = \frac{(-30)(-40)}{-30+40} = 120$cm m $= -\left[\frac{v}{u}\right] = -\left[\frac{120}{-30}\right] = +4$ virtual, erect and magnified

(R) $\frac{u_2}{v} - \frac{u_1}{u} = \frac{u_2 - u_1}{R}$

$\frac{1}{v} - \frac{3/2}{-20} = \frac{1 - 3/2}{-10}$ $\frac{1}{v} + \frac{3}{40} = \frac{1}{20}$

$\frac{1}{v} = \frac{1}{20} - \frac{3}{40} = \frac{2-3}{40} = -\frac{1}{40} \Rightarrow v = -40$

$m = +\frac{u_i}{u_r}\frac{v}{u} = \frac{+3/2}{1} \times \left[\frac{-40}{-20}\right] = +3$

virtual, erect, magnified. (S) V =

$\frac{uF}{u+F} = \frac{(-40)(-40)}{-40-40} = \frac{1600}{-80} = -20$ m $= \frac{v}{u} = \frac{-20}{-40} = +\frac{1}{2}$ virtual, erect, diminished.

PHYSICS

SEC-3

2. Ans (9.00)

$\phi_{net} = \frac{q_{inside}}{\varepsilon_0} = \frac{6\lambda}{\varepsilon_0}$

for $I_{octant} = \frac{6\lambda}{8\varepsilon_0} = \frac{3\lambda}{4\varepsilon_0}$

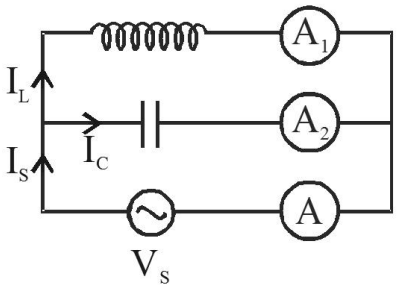
4. Ans (1.00)

$$\frac{p^2}{2m} = \frac{hc}{\lambda} - \phi = \simeq \frac{hc}{\lambda} = \frac{h^2}{2m\lambda^2}$$

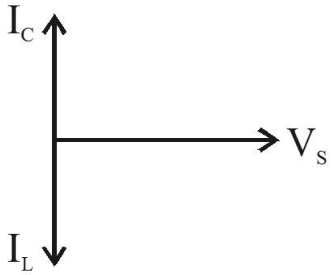
$$\lambda = \frac{h}{2mc} = \frac{6.60 \times 10^{-34}}{2 \times 9.1 \times 10^{-31} \times 3 \times 10^8}$$

$$= \frac{11}{91} \times 10^{-11}\ \text{m} = 1.2\ \text{pm}$$

5. Ans (2.00)



Phasor diagram



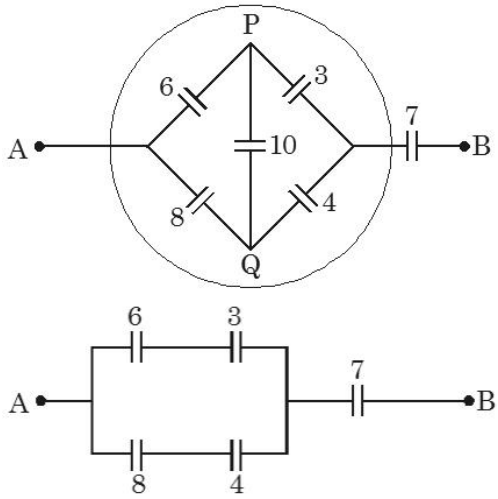
$I_S = I_C + I_L$

$= 5 - 3$

$= 2A$

6. Ans (2.80)

Balanced Wheat stone Bridge



$$C_{eq} = \frac{14}{5}\mu F = 2.80\mu F$$

7. Ans (1.20)

$N = N_0 e^{-\lambda t}$

$N_{decays} = N_0(1 - e^{-\lambda t})$

$= 10^{13}\left(1 - \frac{1}{4}\right) = \frac{3}{4} \times 10^{13}$

$Q = 1.6 \times 10^{-19} \times \frac{3}{4} \times 10^{13} = 1.2\ \mu C$

8. Ans (20.00)

$10^3 \times a^3 \times g = 25\ mg$

$10^3 \times 0.8\ a^2yg + 10^3a^2(a - y)g = mg$

$\Rightarrow a^3g = 25 + a^3g - a^2yg + 0.8\ a^2yg$

$10^3 \times 0.2a^2yg = 25$

$10^3 \times 0.2a^3 \times 10 = 4$

$a^3 = \frac{8}{1000}$

$a = \frac{2}{10} = 0.2\ m$

CHEMISTRY

SEC-3

1. Ans (3.00)

$$\ln\left[\frac{(t_{1/2})_1}{(t_{1/2})_2}\right] = \frac{Ea}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$\ln\left[\frac{3}{(t_{1/2})_2}\right] = \frac{840R}{R}\left(\frac{1}{300} - \frac{1}{400}\right)$$

$(t_{1/2})_2 = 1.5\ \text{min.}$

2. Ans (200.00)

$W = -2(1-2)$

$= 200\ J$

3. Ans (448.00)

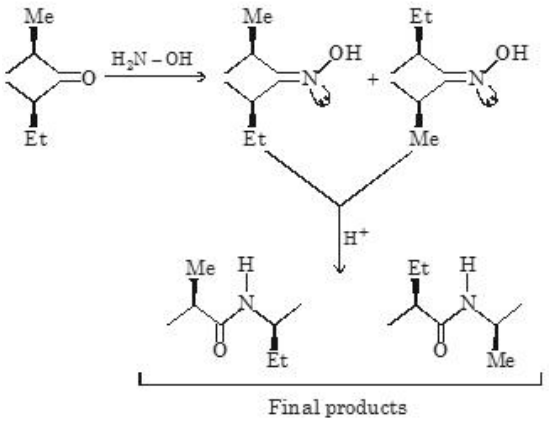
$$(0.4 + n) = \frac{2.8 \times 4}{0.0821 \times 273}$$

$n = 0.1$

$$\frac{0.4}{0.1} = \sqrt{\frac{M}{28}} \Rightarrow M = 448$$

4. Ans (2.00)

The configuration of migratary group remains unchanged



5. Ans (5.00)

(i), (ii), (iii), (vi), (viii)

MATHEMATICS

SEC-1

1. Ans (B,C)

$$\frac{x^4}{4}+\frac{y^2}{9}=1 \Rightarrow e^2=1-\frac{4}{9}=\frac{5}{9}$$

$$\frac{x^4}{4}-\frac{y^2}{9}=1 \Rightarrow e^2=1+\frac{9}{4}=\frac{13}{4}$$

$3 < e_1^2 + e_2^2 < 4$

2. Ans (C,D)

For $x \in [0, 1]$, $f'(x) \geq f'(1)$ (given condition in question)

$$\therefore \frac{f'(1)}{f^2(x)+1} \leqslant \frac{f'(x)}{f^2(x)+1}$$

On integrating we get,

$$f'(1).\int_0^1 \frac{dx}{f^2(x)+1} \leq \tan^{-1}(f(1)) - \tan^{-1}(f(0))$$

i.e. $$\int_0^1 \frac{dx}{f^2(x)+1} \leqslant \frac{\tan^{-1}f(1)}{f'(1)} \leqslant \frac{f(1)}{f'(1)}$$

[$\because \tan^{-1}x \leq x$ for $x \geq 0$]

For equality to hold, $\tan^{-1}f(1) = f(1)$

i.e., $f(1) = 0$ then $$\int_0^1 \frac{dx}{f^2(x)+1} = 0$$

which is not possible as this is a strictly positive function. Hence,

$$\int_0^1 \frac{dx}{f^2(x)+1} < \frac{f(1)}{f'(1)}.$$

3. Ans (B,C,D)

Let O(0,0,0), A(0,0,2), B(0,4,0), C(6,0,0) P is point equidistant from zx, xy and yz plane. Hence its coordinates are P(r,r,r).

Equation of plane ABC is

$2x + 3y + 6z - 12 = 0$

$$r = \left|\frac{11r-12}{\sqrt{49}}\right| \Rightarrow r = 3, \frac{2}{3}$$

since $r < 2 \Rightarrow r = \frac{2}{3}$

4. Ans (A,B)

$$A_k = \begin{bmatrix} C_{k-1} & 0 \\ 0 & C_k \end{bmatrix}, A_{k+1} = \begin{bmatrix} C_k & 0 \\ 0 & C_{k+1} \end{bmatrix}$$

$$A_k A_{k+1} = \begin{bmatrix} C_{k-1}C_k & 0 \\ 0 & C_k C_{k+1} \end{bmatrix}$$

$$\sum_{k=1}^{n-1} A_k A_{k+1} = \begin{bmatrix} \sum_{k=1}^{n-1} C_{k-1}C_k & 0 \\ 0 & \sum_{k=1}^{n-1} C_k C_{k+1} \end{bmatrix}$$

$$a = \sum_{k=1}^{n-1} C_{k-1}C_k = {}^{2n}C_{n-1} - n$$

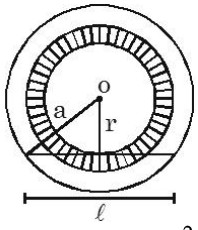
$$b = \sum_{k=1}^{n-1} C_k C_{k+1} = {}^{2n}C_{n+1} - n$$

$a = b$

$a + b = 2({}^{2n}C_{n-1} - n)$

5. Ans (A,B)

Let ℓ be the chord, r be the distance of mid point of chord from the centre of given circle with radius a.



$$\Rightarrow r^2 = a^2 - \frac{\ell^2}{4}$$

$$\ell \in \left(\frac{2a}{a}, \frac{6a}{4}\right), r \in \left(\frac{\sqrt{7}}{4}a, \frac{\sqrt{3}}{2}a\right)$$

The given condition is satisfied if mid point of chord is one of the points lying with in the circles of radii $\frac{\sqrt{7}a}{4}$ and $\frac{\sqrt{3}a}{2}$ concentric with given circles.

The probability

$$= \frac{\text{The area of thin circular ring}}{\text{The area of circle}} = \frac{5}{16}$$

6. Ans (A,D)

The given equation is

$acz_1z_2 + ad\bar{z}_1z_2 + bd\bar{z}_1\bar{z}_2 + bc\bar{z}_1z_2$

$= acz_1z_2 + ad\bar{z}_1z_2 + bcz_1\bar{z}_2 + bd\bar{z}_1\bar{z}_2$

i.e. $(ad - bc)(z_1\bar{z}_2 - \bar{z}_1z_2) = 0$

$\Rightarrow ad - bc = 0 \Rightarrow \frac{a}{b} = \frac{c}{d}$

If $z_1\bar{z}_2 - \bar{z}_1z_2 = 0$

θ_1 and θ_2 are arguments of z_1, z_2

$\Rightarrow \frac{|z_1|e^{i\theta_1}}{|z_1|e^{-i\theta_1}} = \frac{|z_2|e^{i\theta_2}}{|z_2|e^{-i\theta_2}} \Rightarrow \theta_1 = \theta_2$

MATHEMATICS

SEC-2

1. Ans (A)

(P) $\phi(x) = 3f\left(\frac{x^2}{3}\right) + f(3 - y^2)$

$\phi'(x) = 2x\left(f'\left(\frac{x^2}{3}\right) - f'(3 - x^2)\right)$

Now $\frac{x^2}{3} - (3 - y^2) = \frac{4x^2}{3} - 3 > 0 \Rightarrow x < -\frac{3}{2},\ x > \frac{3}{2}$

$\frac{4x^2}{3} - 3 < 0 \Rightarrow -\frac{3}{2} < x < \frac{3}{2}$ $f'\left(\frac{x^2}{3}\right) - f'(3 - x^2)$ is positive in $\left(-3, -\frac{3}{2}\right)$ and $\left(\frac{3}{2}, 4\right)$

and negative in $\left(-\frac{3}{2}, \frac{3}{2}\right)$ as $f'(x)$ is increasing ($\because f''(x) > 0$) $\therefore$ $\phi'(x)$ is negative in

$\left(-3, -\frac{3}{2}\right)$, $\left(0, \frac{3}{2}\right)$ i.e. decreasing in $\left[-3, -\frac{3}{2}\right]$, $\left[0, \frac{3}{2}\right]$ and increasing in

$\left[\frac{3}{2}, 4\right]$, $\left[-\frac{3}{2}, 0\right]$ (Q) $g(x) = (a^2 - 3a + 2)\cos\frac{x}{2} + (a - 1)x + \sin 1$

$f'(x) = -\frac{1}{2}(a^2 - 3a + 2)\sin\frac{x}{2} + (a - 1)$

$= (a - 1)\left(1 - \frac{1}{2}(a - 2)\sin\frac{x}{2}\right)$

$f'(x) = 0$ when $a = 1$ or $\sin\frac{x}{2} = \frac{2}{a - 2}$, $a \neq 2$ $\therefore \left|\frac{2}{a - 2}\right| < 1 \Rightarrow a \in$ $(-\infty, 0) \cup (4, \infty)$

$\Rightarrow$ It possess critical points

if $a \in (-\infty, 0] \cup [4, \infty) \cup \{1\}$

2. Ans (B)

$S : x^2 + y^2 - 2x - 4y + 1 = 0,$

centre (1, 2), radius = 2

$S' : x^2 + y^2 - 4x - 2y + 1 = 0,$

centre (2, 1), radius = 2

(P) (1, 3) lies inside S and outside S'.

$\Rightarrow$ Sum of tangents is 2.

(Q) Required circle is

$x^2 + y^2 - 2x - 4y + 1 + \lambda(2x - 2y) = 0$

$x^2 + y^2 - 2(1 - \lambda)x - 2(2 + \lambda)y + 1 = 0$

centre is $(1 - \lambda, 2 + \lambda)$

it lies on $2x + y - 6 = 0$

$\Rightarrow 2 - 2\lambda + 2 + \lambda - 6 = 0$

$-\lambda - 2 = 0 \Rightarrow \lambda = -2$

radius $= \sqrt{9 + 0 - 1} = \sqrt{8}$

$\frac{r^2}{4} = 2$

(R) $m = -1$

(S) Circles are intersecting

$\Rightarrow$ length of tangent

$= \sqrt{(C_1C_2)^2 - (r_1 - r_2)^2} = 2$

3. Ans (A)

(P) ${}^{39}C_{3r-1} - {}^{39}C_{r^2} = {}^{39}C_{r^2-1} - {}^{39}C_{3r}\ {}^{40}C_{3r} = {}^{40}C_{r^2}$

$\Rightarrow$ (1) $r = 3$ (2) $r^2 + 3r - 40 = 0 \Rightarrow r = 5$

(Q) ${}^{2n+3}C_{2n} - {}^{2n+2}C_{2n-1} = 15(2n + 1)\ {}^{2n+2}C_{2n} = 15(2n + 1)$

$(2n + 2)(2n + 1) = 30(2n + 1) \Rightarrow n = 14$

(R) ${}^{56}P_{r+6} : {}^{54}P_{r+3} = 30800 : 1\ \frac{56!}{(50 - r)!} \times \frac{(51 - r)!}{54!} = \frac{30800}{1}$

$56 \times 55 \times (51 - r) = 30800 \quad 51 - r = 10 \Rightarrow r = 41$

(S) $\binom{n+2}{8} : {}^{n-2}P_4 = 54 : 16\ \frac{(n + 2)!}{8!(n - 6)!} \times \frac{(n - 6)!}{(n - 2)!} = \frac{57}{16}$

$\frac{(n + 2)(n + 1)(n)(n - 1)}{8!} = \frac{57}{16}\ (n + 2)(n + 1)(n)(n - 1) = 143640$

$= 18 \times 19 \times 20 \times 21 \Rightarrow n = 19$

4. Ans (C)

(P) H.M. $= 4 \Rightarrow \frac{2ab}{a + b} = 4 \Rightarrow ab = 2(a + b)$ ...(1)

$A = \frac{a + b}{2}, G^2 = ab, 2A + G^2 = 27 \Rightarrow a + b + ab = 27$ (1), (2) $\Rightarrow$

$a + b = 9,$

$ab = 18 \Rightarrow a = 3, b = 6 \Rightarrow a^2 + b^2 = 45$ (Q) a, x, y, z, b are in A.P. $\Rightarrow y = \frac{a + b}{2}$

$x + z = a + d + b - d = a + b$, where d is the common difference.

$\therefore x + y + z = 15 \Rightarrow \frac{a + b}{2} + (a + b) = 15\ a + b = 10$ ...(1)

a, x, y, z, b are in H.P.

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{1}{2}\left(\frac{1}{a} + \frac{1}{b}\right) + \left(\frac{1}{a} + \frac{1}{b}\right)$$

$$= \frac{3}{2}\left(\frac{1}{a} + \frac{1}{b}\right)$$

$$= \frac{3}{2}\frac{(a + b)}{ab} = \frac{5}{8}, \text{given } \therefore$$

$$ab = \frac{8}{5} \times \frac{3}{2} \times 10 = 24 \text{ [using (1)]} \quad ...(2)$$

Using (1) and (2) $\Rightarrow$ (a, b) = (4, 6) $\Rightarrow a^2 + b^2 = 52$

$$(R) \sum_{r=1}^{16} \left(\frac{1^3 + 2^3 + \ldots + r^3}{1 + 3 + \ldots + (2r - 1)}\right)$$

$$= \sum_{1}^{16} \frac{1}{4}(r + 1)^2 = \frac{1}{4}(2^2 + 3^2 + \ldots + 17^2)$$

$$= \frac{1}{4}\left(\frac{17 \times 18 \times 35}{6} - 1\right) = 446 \text{(S) } G^2 =$$

AH $18^2 = A.\frac{216}{13} \Rightarrow A = \frac{39}{2}$

x_1 and x_2 are roots of $x^2 - 2Ax + G^2 = 0$

$$|x_1 - x_2| = \sqrt{(x_1 + x_2)^2 - 4x_1x_2}$$

$$= \sqrt{4A^2 - 4G^2} = 2\sqrt{A^2 - G^2} = 15$$

MATHEMATICS

SEC-3

1. Ans (1.20)

$$\tan\left(\frac{A-B}{2}\right)=\frac{a-b}{a+b}\cot\left(\frac{C}{2}\right)=\frac{1}{9}\cot\frac{C}{2}$$

$$\frac{31}{32}=\cos(A-B)$$

$$=\frac{1-\tan^2\left(\frac{A-B}{2}\right)}{1+\tan^2\left(\frac{A-B}{2}\right)}=\frac{1-\frac{1}{81}\cot\frac{C}{2}}{1+\frac{1}{81}\cot^2\frac{C}{2}}$$

$$\cot^2\left(\frac{C}{2}\right)=\frac{81}{63}\Rightarrow\tan^2\frac{C}{2}=\frac{7}{9}$$

$$\Rightarrow\cos C=\frac{1}{8}$$

$$\frac{a^2+b^2-c^2}{2ab}=\frac{1}{8}\Rightarrow c^2=36$$

2. Ans (7.50)

$(1-x)^{30}(1+x+x^2)^{31}$

$=(1-x^3)^{30}(1+x+x^2)$

Coefficient of x^{47} = coeff. of x^{45} in $(1-x^3)^{30}$

$=-{}^{30}C_{15}$

3. Ans (0.25)

Sum of the coefficients = 0

$\Rightarrow$ one root is 1

$x^5-5x^4+9x^3-9x^2+5x-1=0$

$(x+1)(x^4-4x^3+5x^2-4x+1)=0$

$$x^2-4x+5-\frac{4}{x}+\frac{1}{x^2}=0$$

$$\left(x+\frac{1}{x}\right)^2-2-4\left(x+\frac{1}{x}\right)+5=0$$

$$x+\frac{1}{x}=1,\ 3$$

(1) $x+\frac{1}{x}=3\Rightarrow x^2-3x+1=0$

$$x=\frac{3\pm\sqrt{5}}{2}x=\frac{3\pm\sqrt{5}}{2}$$

(2) $x+\frac{1}{x}=1x+\frac{1}{x}=1$ (rejected)

real roots are $1,\ \frac{3+\sqrt{5}}{2},\ \frac{3-\sqrt{5}}{2}$

$1,\ \frac{3+\sqrt{5}}{2},\ \frac{3-\sqrt{5}}{2}$

4. Ans (0.40)

Let $x=\frac{1}{t}x=\frac{1}{t}$

$$\lim_{t\to0}\left(\frac{f(3+3t)}{f(3)}\right)^{\frac{1}{t}}=e^{\lim_{t\to0}\left(\frac{f(3+3t)-f(3)}{tf(3)}\right)}$$

$$\lim_{t\to0}\left(\frac{f(3+3t)}{f(3)}\right)^{\frac{1}{t}}=e^{\lim_{t\to0}\left(\frac{f(3+3t)-f(3)}{tf(3)}\right)}$$

$$=e^{\lim_{t\to0}\left(\frac{f'(3+3t).3}{f(3)}\right)}=e^{\lim_{t\to0}\left(\frac{f'(3+3t).3}{f(3)}\right)}=e$$

5. Ans (0.42, 0.43)

$$I=\int_{\pi/4}^{3\pi/4}\frac{\cos x}{1-\cos x}\,dxI=\int_{\pi/4}^{3\pi/4}\frac{\cos x}{1-\cos x}\,dx$$

(k and A)

$$2I=\int_{\pi/4}^{3\pi/4}\cos x\left(\frac{1}{1-\cos x}-\frac{1}{1+\cos x}\right)dx$$

$$2I=\int_{\pi/4}^{3\pi/4}\cos x\left(\frac{1}{1-\cos x}-\frac{1}{1+\cos x}\right)dx$$

$$2I=2\int_{\pi/4}^{3\pi/4}\cot^2x\,dx=2\left(2-\frac{\pi}{2}\right)$$

$$2I=2\int_{\pi/4}^{3\pi/4}\cot^2x\,dx=2\left(2-\frac{\pi}{2}\right)$$

$$I=2-\frac{\pi}{2}I=2-\frac{\pi}{2}$$

6. Ans (1.45)

$$f(x)=\begin{vmatrix}\sec x & \cos x & \sec^2x+\cot x\,\mathrm{cosec}x+\cos x\\ \cos^2x & \cos^2x & \mathrm{cosec}^2x+\cos^4x\\ 1 & \cos^2x & \cos^2x\end{vmatrix}$$

$(R_1\to R_1-\sec xR_3,\ R_2\to R_2-\cos^2xR_3)$

$$f(x)=\begin{vmatrix}0 & 0 & \sec^2x+\cot x\,\mathrm{cosec}x\\ 0 & \cos^2x-\cos^4x & \mathrm{cosec}^2x\\ 1 & \cos^2x & \cos^2x\end{vmatrix}$$

$=(-\cos^2x+\cos^4x)(\sec^2x+\cot x\,\mathrm{cosec}x)$

$=-\sin^2x-\cos^3x$

$$\int_0^{\pi/2}f(x)\,dx=\int_0^{\pi/2}(-\sin^2x-\cos^2x)\,dx$$

$$\int_0^{\pi/2}f(x)\,dx=\int_0^{\pi/2}(-\sin^2x-\cos^2x)\,dx$$

$$=-\left(\frac{\pi}{4}+\frac{2}{3}\right)=-\left(\frac{\pi}{4}+\frac{2}{3}\right)$$

7. Ans (4.66, 4.67)

x-coordinate of point of intersection is given by

$4x^2 - 4(3 + a)x + 9 + \frac{4a}{3}\frac{4a}{3} = 0$

$D = 0$

$(3+a)^2 - \left(9 + \frac{4a}{3}\right) = 0$

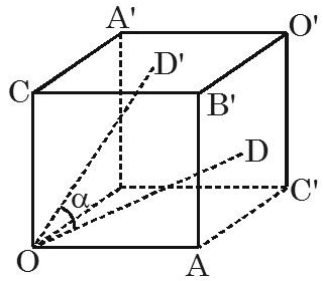
$(3+a)^2 - \left(9 + \frac{4a}{3}\right) = 0$

$a^2 + 6a - \frac{4a}{3}\frac{4a}{3} = 0$

$a^2 + \frac{14a}{3} = 0 a^2 + \frac{14a}{3} = 0 \Rightarrow a = 0, -\frac{14}{3}$

$-\frac{14}{3}$

8. Ans (4.50)



Let $\left|\overrightarrow{OA}\right| = 2\left|\overrightarrow{OA}\right| = 2$

$\left|\overrightarrow{OB}\right| = 4\left|\overrightarrow{OB}\right| = 4$

$\left|\overrightarrow{OC}\right| = 6\left|\overrightarrow{OC}\right| = 6$

$\overrightarrow{OD} = \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AC'} + \frac{1}{2}\overrightarrow{AB'}$

$\overrightarrow{OD} = \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AC'} + \frac{1}{2}\overrightarrow{AB'}$

$= 2\hat{i} + 2\hat{j} + 3\hat{k} = 2\hat{i} + 2\hat{j} + 3\hat{k}$

$\overrightarrow{OD'} = \overrightarrow{OC} + \frac{1}{2}\overrightarrow{CB'} + \frac{1}{2}\overrightarrow{B'O'}$

$\overrightarrow{OD'} = \overrightarrow{OC} + \frac{1}{2}\overrightarrow{CB'} + \frac{1}{2}\overrightarrow{B'O'}$

$= \hat{i} + 2\hat{j} + 6\hat{k} = \hat{i} + 2\hat{j} + 6\hat{k}$

$\cos\alpha = \frac{\overrightarrow{OD}.\overrightarrow{OD'}}{\left|\overrightarrow{OD}\right|\left|\overrightarrow{OD'}\right|} = \frac{24}{\sqrt{17.41}}$

$\cos\alpha = \frac{\overrightarrow{OD}.\overrightarrow{OD'}}{\left|\overrightarrow{OD}\right|\left|\overrightarrow{OD'}\right|} = \frac{24}{\sqrt{17.41}}$

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2020 - 2021)

JEE (Advanced)
MAJOR
13-05-2021

JEE(Main + Advance) : (PHASE : ALL PHASES)

Test Type : MAJOR Test Pattern : JEE (Advanced)

TEST DATE : 13-05-2021

ANSWER KEY PAPER-2

PHYSICS

SEC-1	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C,D	A,B,C	A,B,C,D	A,B	A,B,C	B,D	A,B	A,B
SEC-2	Q.	1	2	3	4				
	A.	D	A	B	A,B,C				
SEC-3	Q.	1	2	3	4	5	6	7	8
	A.	3	1	5	6	3	2	8	4

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C	A,B	A,B,C	A,B,D	A,C,D	B,D	A,B,D	A,C
SEC-2	Q.	1	2	3	4				
	A.	C	A	A,C	A,B,D				
SEC-3	Q.	1	2	3	4	5	6	7	8
	A.	9	3	2	4	1	6	3	6

MATHEMATICS

SEC-1	Q.	1	2	3	4	5	6	7	8
	A.	A,C	A,C	A,C,D	A,B,C	A,B	B,D	A,B	B,C,D
SEC-2	Q.	1	2	3	4				
	A.	B	C	A,B,C	A				
SEC-3	Q.	1	2	3	4	5	6	7	8
	A.	7	2	2	1	8	6	4	1

HINT – SHEET

PHYSICS

SEC-1

1. **Ans (A,B,C,D)**

(A) $\vec{F}_{net} = \vec{O} \Rightarrow$ particle is in equilibrium

(B) Bottom most point is AOR, so no point has vertical velocity

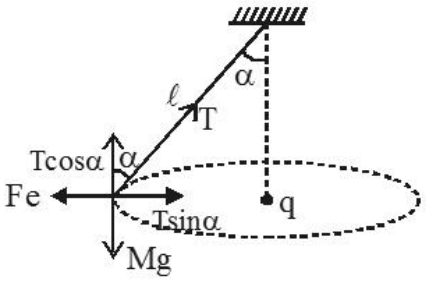
(C) Because of angular momentum conservation, as 'I' increases 'w' must decrease.

(D) The resultant of centripetal force and tangential force will be skew.

2. Ans (A,B,C)

$T\cos\alpha = Mg$

$T\sin\alpha - Fe = \frac{Mv^2}{r}$



3. Ans (A,B,C,D)

In standing wave energy is confined between node & antinode.

Node and antinode do not transfer its energy but element between node & antinode transfers its energy to other element.

4. Ans (A,B)

$(A)\ \lambda_i = \frac{10u - \left(\frac{u}{2}\right)}{f} = \frac{19u}{2f}$

$(B)\ f_i = \frac{10u - u}{10u - \frac{u}{2}} . f = \frac{18f}{19}$

$(C)\ \lambda_r = \frac{10u + u}{f_r} = \frac{11u \times 19}{18f} = \frac{11 \times 19u}{18f}$

$(D)\ f_r = \frac{18f}{19}$

5. Ans (A,B,C)

$TV^{\gamma-1}$ = constant

$\Rightarrow V^{\gamma-1}\ 1 + T\ (\gamma - 1)\ V^{\gamma-2}\frac{dV}{dt} = 0$

$\Rightarrow \frac{dV}{dT} = -\frac{V}{T\ (\gamma - 1)}$

$\Rightarrow \left|\frac{dV}{dt}\right|_{T_0,V_0} = \frac{V_0}{T_0(\gamma - 1)} = m \Rightarrow$

$\gamma - 1 = \frac{V_0}{T_0 m}$

$\Rightarrow C_v = \frac{R}{\gamma - 1} = \frac{Rt_0 m}{V_0} \Rightarrow C_p = C_v + 1 =$

$\left(1 + \frac{T_0 m}{V_0}\right) R$

6. Ans (B,D)

Use Ampere's circuital law.

7. Ans (A,B)

$\frac{R}{R_1 + R_2} = \frac{L/4}{3L/4}$

$3R = (R_1 + R_2)$

$\frac{R + R_1}{R_2} = \frac{2/3}{1/3}$

$3R = R_2 + R_1$

$R = 2R_2 - R_1$

$4R = 3R_2$

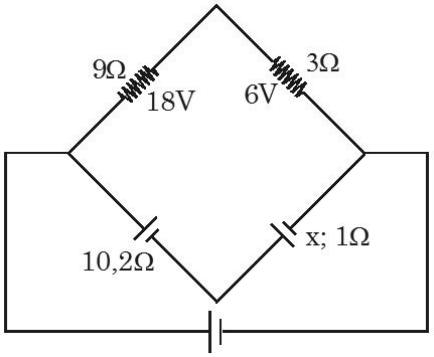
$R_1 = \frac{5R}{3}$

$R_2 = \frac{4R}{3}$

PHYSICS

SEC-2

1. Ans (D)



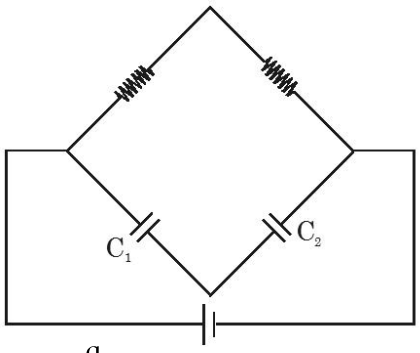
$18 = 10 + 2i$

$i = 4A$

$x + 4 = 6$

$x = 2$ volt

2. Ans (A)



$V_1 = \frac{q}{C_1} = 18$

$V_2 = \frac{q}{C_2} = 6$

$\Rightarrow \frac{C_1}{C_2} = \frac{1}{3}$

4. Ans (A,B,C)

Particle would travel in cycloidal path.

$x = vt - R\cos\omega t$

$y = R(1 - \cos\omega t)$

where, $v = \frac{E_0}{B_0}, \omega = \frac{qB_0}{m}, R = \frac{mE_0}{qB_0^2}$

$y_{max} = 2R = \frac{2mE_0}{qB_0^2}$

and radius of curvature at y_{max}

$= 4R = \frac{4mE_0}{qB_0^2}$

PHYSICS

SEC-3

1. Ans (3)

$T \propto \sqrt{\text{length}}$ free length becomes $\frac{\ell}{4}$ So time period becomes $T_1 = \frac{T}{2}$

Half oscillation is with free length ℓ and half with free length $\frac{\ell}{4}$

Total time $= \frac{T}{2} + \frac{T_1}{2} = \frac{3T}{4}$

2. Ans (1)

$\Delta x = (\mu_{rel} - 1)\, t = \left[\frac{\mu_2}{\mu_1} - 1\right] t$

$\mu_1 = \frac{4}{3}$

$\mu_2 = \frac{3}{2}$

$t = 8\ \mu m$

$\Delta x = 1\mu m$

3. Ans (5)

Temp. is increased by $\Delta\theta$ then

$\Delta\ell = \ell\alpha\Delta\theta$

$\Rightarrow \Delta\theta = \frac{\Delta\ell}{\ell\alpha}$

$E_1 = (\rho A\ell)S\Delta\theta = \rho A\ell\, S\frac{\Delta\ell}{\ell\alpha}$

when stretched, Stress $= Y\frac{\Delta\ell}{\ell}$

$E_2 = \frac{1}{2}\left(Y\frac{\Delta\ell}{\ell}\right)\left(\frac{\Delta\ell}{\ell}\right) \times A\ell = \frac{Y(\Delta\ell)^2 A}{2\ell}$

So, $\frac{E_1}{E_2} = \frac{E_1}{E_2} = \frac{\rho A\ell S\Delta\ell \times 2\ell}{\ell \times Y(\Delta\ell)^2 A} = \frac{2\rho S\ell}{\alpha(\Delta\ell)Y}$

$= 500$

5. Ans (3)

For image formed by lens

$\frac{1}{v_1} - \frac{1}{-15} = \frac{1}{10} \Rightarrow v_1 = +30cm$

i.e. 20 cm behind mirror For mirror

$\frac{1}{v_2} + \frac{1}{20} = \frac{1}{-20} \Rightarrow v_2 = -10$ cm

Overall magnification =

$\left(\frac{30}{-15}\right) \times \left(\frac{10}{20}\right) = 1$

Length of image $= 1 \times 3 = 3$ mm

6. **Ans (2)**

$d\phi = (2\pi x dx)kxt^2$

$\phi = \frac{2\pi}{3}kx^3t^2$

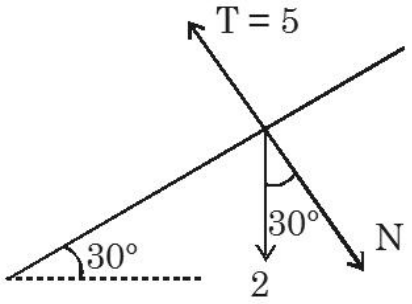
$E = 2\pi x = \frac{4\pi kx^3t}{3}$

$E = \frac{2ktx^2}{3}$

$d\tau = \left(\frac{Q}{\pi R^2}2\pi x dx\right)\frac{2}{3}ktx^2x$

$\tau = \frac{4ktQ}{3R^2}\int_0^R x^4dx = \frac{4ktQR^3}{15}$

7. **Ans (8)**

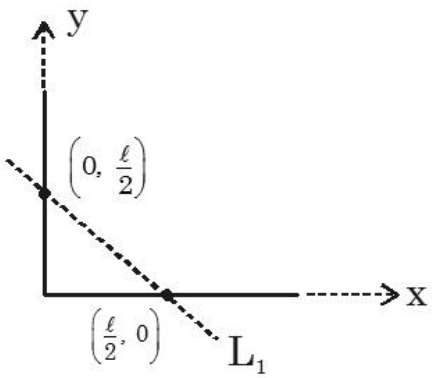


$N + \sqrt{3} = 5$

$f_\ell = \mu N$

$\Rightarrow 1 = \mu(5 - \sqrt{3}) \Rightarrow \mu = \frac{1}{5 - \sqrt{3}}$

8. **Ans (4)**



Centre of mass must lie on line L_1

$\frac{x}{\frac{\ell}{2}} + \frac{y}{\frac{\ell}{2}} = 1$

$2x + 2y = \ell$

$a + b = 4$

CHEMISTRY

SEC-1

1. Ans (A,B,C)

$$CO_3^{2-} + H_2O \rightleftharpoons HCO_3^{-} + OH^{-}, \quad K_{h_1} = \frac{10^{-14}}{10^{-9}} = 10^{-5}$$
(0.1−x) (x−y) (x+y)

$$HCO_3^{-} + H_2O \rightleftharpoons H_2CO_3 + OH^{-}, \quad K_{h_2} = \frac{10^{-14}}{10^{-5}} = 10^{-9}$$
(x−y) (y) (x+y)

Neglect y w.r.t. x, then

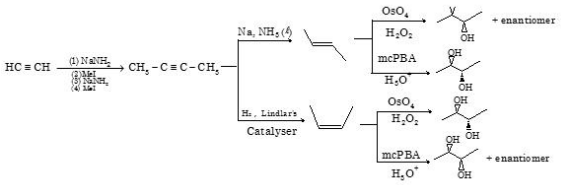
$$K_{h_1} = 10^{-5} = \frac{x^2}{0.1-x} \approx \frac{x^2}{0.1} \Rightarrow x \approx 10^{-3}$$

$$y = K_{h_2} = 10^{-9} = [H_2CO_3]$$

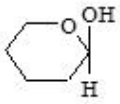
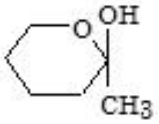
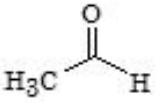
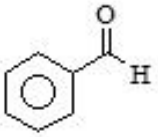
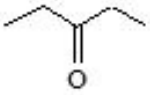
$$[OH^-] \approx x = 10^{-3}M \Rightarrow pOH = 3,\ pH = 11$$

$$[Co_3^{2-}] \approx 0.1M$$

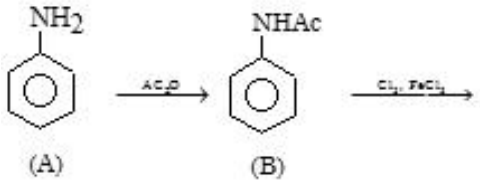
3. Ans (A,B,C)

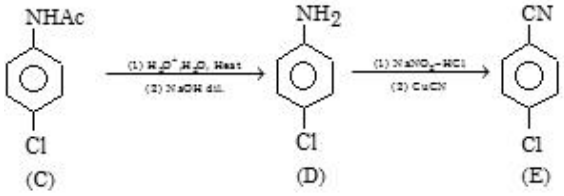


4. Ans (A,B,D)

Comp.	Fehling Test	Tollen's test
HCHO	Y	Y
	N	N
	Y	Y
	N	N
	Y	Y
	N	Y
	N	N

5. Ans (A,C,D)





.

8. Ans (A,C)

$Fe + dilH2SO_4 \rightarrow FeSO_4 + H_2$

CHEMISTRY

SEC-2

1. Ans (C)

At 27°C

$E_{cell} = 0.4 - 2 \times 10^{-6} \ (300)2 = 0.4 - 0.18 =$ 0.22 V

$\Delta G = -0.22$ F

2. Ans (A)

$$\left[\frac{dE_{cell}}{dT}\right]_P = -4 \times 10^{-6} \times T = -1.2 \times 10^{-3} \text{ V/K}$$

$\Delta S = 1 \times F \times (-1.2 \times 10^{-3}) = -115.8$ J/K

3. Ans (A,C)

$4BF_3 + 3H_2O \rightarrow H_3BO_3 + 3HBF_4$

$SiCl_4 + 4H_2O \rightarrow H_4SiO_4 + 4\ HCl$

$3SiF_4 + 4H_2O \rightarrow H_4SiO_4 + 2H_2SiF_6$

$SOCl_2 + H_2O \rightleftharpoons H_2SO_3 + 2HCl$

4. Ans (A,B,D)

$PCl_3 + 3H_2O \rightarrow H_3PO_3 + 3HCl$

$PCl_5 + H_2O \rightarrow POCl_2 + 2HCl$

$2NCl_3 + 6H_2O \rightarrow 2NH_3 + 6HClO$

CHEMISTRY

SEC-3

1. Ans (9)

Initial mass ratio of H_2 & He = 1 : 4

$\Rightarrow$ Initial molar ratio of H_2 & He = 1 : 2

Final molar ratio of H_2 and He = 8 : 1

we have,

$$\left(\frac{n_{H_2}}{n_{H_e}}\right)_{n-step} = \left(\frac{n_{H_2}}{n_{H_e}}\right)_{initial} \left(\sqrt{\frac{M_{H_e}}{M_{H_2}}}\right)^n$$

$$\frac{8}{1} = \frac{1}{2} \cdot \left(\sqrt{\frac{4}{2}}\right)^n \Rightarrow n = 8$$

$\Rightarrow$ minimum number of containers required = n + 1 = 9

2. Ans (3)

G = H – TS

As process is reversible adiabatic the entropy stays constant.

$\Rightarrow$ dG = dH – SdT

$$\Delta G = \Delta H - S\Delta T = nC_{P,m}\Delta T - S\Delta T = 1 \times \frac{7}{2} \times R\Delta T - S\Delta T$$

$$= \frac{7}{2} \times 2 \times \Delta T - 3\Delta T = 4 \times \Delta T$$

for adiabatic process

$$T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$$

$$T_2 = T_1 \times \left(\frac{V_1}{V_2}\right)^{\gamma-1} = 1000 \times \left(\frac{1}{32}\right)^{7/5-1} = 250\,K$$

$$\Delta T = 250 - 1000 = -750$$

$$\Rightarrow \Delta G = 4 \times 4T = 4 \times -750 = -3000 \text{ cal}$$

$$\Rightarrow |\Delta G| = 3\,kcal$$

3. Ans (2)

$$\frac{d}{dT}(\ell nK) = \frac{E_a}{RT^2}$$

$$\int d(\ell n\, K) = \int \frac{E_a}{RT^2}.dT$$

$$\ell nK = \int \frac{0.01T^2 + \frac{20}{7}T}{RT^2}.dT$$

$$\ell nk = \frac{0.01}{R}.T + \frac{20}{7R}\ell nT + C$$

$$\Rightarrow \ell n\frac{k_2}{k_1} = \frac{0.01}{R}(400 - 200) + \frac{20}{7R}\ell n\frac{400}{200}$$

$= 1 + 1 = 2$

4. Ans (4)

$$R - CH(OH) + CH_3COCl \rightarrow R - CH - O - \overset{O}{\overset{\|}{C}} - CH_3$$

for every acetylation reaction, molecular mass increases by 42

$$\text{No. of OH group} = \frac{318 - 150}{42} = 4$$

5. Ans (1)

Only (f)

6. Ans (6)

$8 - 2 = 6$

7. Ans (3)

SO_2, SO_3, $N(SiH_3)_3$

8. Ans (6)

$PbCrO_4$, $BaCrO_4$, $(NH_4)_3PO_4.12MoO_3$, $K_3[Co(NO_2)_6]AgI$, PbI_2

MATHEMATICS

SEC-1

1. Ans (A,C)

$f(x)$ =

$1-(\sin 2x)\sqrt[4]{2\left(\tan^2 x+3\right)+\tan^4 x-5}$

$=1-(\sin 2x)\sqrt[4]{\tan^4 x+2\tan^2 x+1}$

$=1-(\sin 2x)\sqrt[4]{\left(\tan^2 x+1\right)^2}$

$=1-(\sin 2x)\sec x$

$=1-2\sin x \in [1,3)$

2. Ans (A,C)

tangents are perpendicular

so they intersect on M(2, 0) lie on directrix

mid point of A(1, 1) and B(4, 2) is $\left(\frac{5}{2},\frac{3}{2}\right)$

slope of axis is $=\frac{\frac{3}{2}-0}{\frac{5}{2}-2}=3$

equation of directrix, $y-0=-\frac{1}{3}(x-2)$

$\Rightarrow 3y+x=2$

AM = AS (S is the focus and M is the foot of perpendicular from A) $\Rightarrow \frac{2}{\sqrt{10}}$

BN = BS (S is the focus and N is the foot of perpendicular from B) $\Rightarrow \frac{8}{\sqrt{10}}$

AS : BS = 1 : 4

$\Rightarrow S\left(\frac{1\times 4+4\times 1}{5},\frac{1\times 2+4\times 1}{5}\right)=\left(\frac{8}{5},\frac{6}{5}\right)$

solve equation of directrix and eq. of axis to get foot of directrix. vertex is mid point of foot of directrix and focus.

3. **Ans (A,C,D)**

$${}^{2n}C_n\left(\frac{1}{2}\right)^n\left(\frac{1}{2}\right)^n=\sum_{r=0}^{n}\left(\frac{{}^nC_r}{2^n}\right)^2= \quad \text{(C)}$$

option

$${}^{2n}C_n\left(\frac{1}{2}\right)^n\left(\frac{1}{2}\right)^n=\frac{2n!}{n!n!2^{2n}}$$

$$=\frac{2^n.\,n!1.3.5....(2n-1)}{n!n!2^{2n}}=\frac{1.3.5.....(2n-1)}{n!2^n}$$

$$\prod_{r=1}^{n}\left(\frac{2r-1}{2r}\right)=\text{(A) option}$$

$$\prod_{r=1}^{n}\left(\frac{n+r}{2r}\right)=\frac{(n+1)(n+2)(n+3)....(2n)}{2.4.6.....2n}$$

$$=\frac{2n!}{n!2^n.\,n!}=\frac{{}^{2n}C_n}{2^n}$$

= (B) option wrong

$$\frac{\sum_{r=0}^{n}({}^nC_r)^2}{\left(\sum_{r=0}^{2n}{}^{2n}C_r\right)}=\frac{({}^nC_0)^2+({}^nC_1)^2+.....+({}^nC_n)^2}{{}^{2n}C_0+.....+{}^{2n}C_n}=\frac{{}^{2n}C_n}{2^{2n}}$$

= option (D)

4. **Ans (A,B,C)**

$$P\left(2^2.5^2\right)=2^2.5^2\left(1-\frac{1}{2}\right)\left(1-\frac{1}{5}\right)=2.5.4$$

$\because$ no. of numbers multiple of $2=2.5^2$

no. of numbers multiple of $5=2^2.5$

no. of numbers multiple of both $=2.5$

$$\therefore P\left(2^2.5^2\right)=2^2.5^2-2.5^2-2^2.5+2.5$$

$$=2^25^2\left(1-\frac{1}{2}\right)\left(1-\frac{1}{5}\right)$$

Similarly check all options.

5. Ans (A,B)

Let the equation of circle 's' is given by

$(x-1)^2+(y-1)^2+\lambda(x+y-2)=0$ ... (1)

sine circle 'x' is orthogonal to the circle

$x^2+y^2+2x+2y-2=0$ then $\lambda=1$

$\Rightarrow$ equation of 's' is $x^2+y^2-x-y=0$

$\Rightarrow$ now the length of the tangent from (2, 2) is $\sqrt{s}=2$.

and the equation of the director circle is

$2x^2+2y^2-2x-2y-1=0$.

6. Ans (B,D)

$$I_m=\int_0^{m\pi} x\,|\sin x|\,e^{\cos 4x}\,dx \quad \text{.... (1)}$$

by using the property

$$\int_0^a f(x)dx=\int_0^q f(a-x)\,dx$$

$$I_m=\int_0^{m\pi}(m\pi-x)\,|\sin x|\,e^{\cos 4x}\,dx \quad \text{....(2)}$$

adding eq. (1) & eq. (2) we get

$$2I_m=m\pi\int_0^{m\pi}|\sin x|\,e^{\cos 4x}\,dx$$

by using periodic property

$$2I_m=m^2\pi\int_0^{\pi}|\sin x|\,e^{\cos 4x}\,dx$$

$$2I_m=2m^2\pi\int_0^{\pi/2}\sin x\,e^{\cos 4x}\,dx$$

$$\Rightarrow I_m=m^2\pi\int_0^{\pi/2}\sin x\,e^{\cos 4x}\,dx \quad \text{...(3)}$$

and

$$J_n=n\int_0^{\pi}|\cos x|\,e^{\cos 4x}\,dx=2n\int_0^{\pi/2}\sin x e^{\cos 4x}\,dx$$

.... (4)

on dividing eq. (3) and (4)

$\frac{I_m}{J_n}=\frac{m^2\pi}{2n}$ now B,D is correct

7. **Ans (A,B)**

Since $f(x) = \cos x \sin 2x = 2\sin x - 2\sin^3 x$

so $\min f(x) = \min g(t)$ where

$g(t) = 2t - 2t^3$ where $t \in [-1, 1]$

Now for maxima/minima, we have

$g'(t) = 2 - 6t^2 = 0 \Rightarrow t = \pm\frac{1}{\sqrt{3}}$

Again g\dblquote $(t) = -12t$

$\therefore$ g\dblquote $\left(\frac{1}{\sqrt{3}}\right) < 0$ & $g'\left(\frac{-1}{\sqrt{3}}\right) > 0$

hence

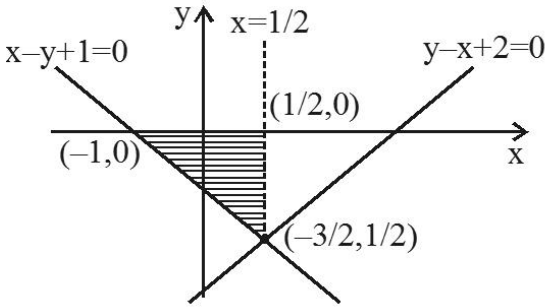
$\min g(t) = g\left(\frac{-1}{\sqrt{3}}\right) = -\frac{4}{3\sqrt{3}} > \frac{-7}{9} > \frac{-9}{7}$

8. **Ans (B,C,D)**

Domain $x + y + 1 > 0$,

$y - x + 2 > 0$

Squring $x + y + 1 < y - x + 2$



i.e.$x < \frac{1}{2}$

Shaded region indicated in the figure

and its area $= \frac{1}{2}\cdot\frac{3}{2}\cdot\frac{3}{2} = \frac{9}{8}$

MATHEMATICS

SEC-2

1. Ans (B)

$(AB + BA)^T = (AB)^T + (BA)^T$

$= B^TA^T + A^TB^T = BA + AB$

$(AB - BA)^T = (AB)^T - (BA)^T$

$= B^TA^T - A^TB^T = BA - AB = -(AB - BA)$

2. Ans (C)

Since A is skew-symmetric matrix therefore |A| = 0

3. Ans (A,B,C)

Using the concept of family of planes let required equation of plane is

$p_1 + \lambda p_2 = 0$, since the plane passes through $(3, 2, 1)$ therefore

$\Rightarrow 2x - y + z - 2 + \lambda(x + 2y - z - 3)$

$\Rightarrow \lambda = -1$,

required plane is x– 3y + 2z + 1=0

4. Ans (A)

Equation of the plane passes through $(-1, 3, 2)$

is $a(x + 1) + b(y - 3) + c(z - 2) = 0.....(1)$

(1) is perpendicular to L = 0,

(Dr's of L are $(2i - j + k) \times \left(i + 2j - \hat{k}\right)$) we get

$\frac{a}{-1} = \frac{b}{3} = \frac{c}{5}$ therefore equation of plane is

$(\vec{r} - (-\hat{i} + 3\hat{j} + 2\hat{k}))\cdot(\hat{i} - 3\hat{j} - 5\hat{k}) = 0$

MATHEMATICS

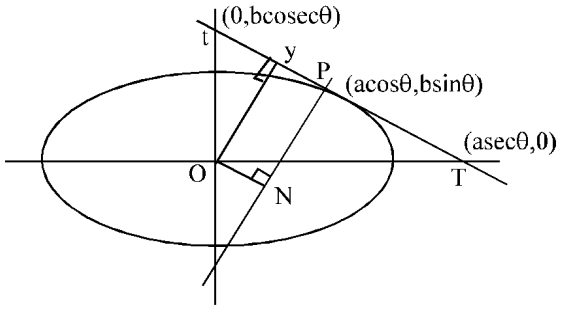
SEC-3

1. Ans (7)

Tangent at P : $\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$

T : $(a\sec\theta , 0)$, t : $(0 , b\,\text{cosec}\theta)$

hence $Tt = \sqrt{a^2\sec^2\theta + b^2\,\text{cosec}^2\theta}$



To find Py, we draw normal at P.

now ON = Py

Hence Normal at P

$\frac{ax}{\cos\theta} - \frac{by}{\sin\theta} = a^2 - b^2$

ON : $\frac{a^2 - b^2}{\sqrt{a^2\sec^2\theta + b^2\,\text{cosec}^2\theta}} = Py$

Hence Py. $Tt = a^2 - b^2$]

2. Ans (2)

$\frac{f(x)}{1+x^2} = 1 + \int_0^x \frac{f^2(t)}{1+t^2}\,dt \Rightarrow f(0) = 1$

differentiate w.r.t. 'x' :

$\frac{(1+x^2)\,.\,f'(x) - 2x\,f(x)}{(1+x^2)^2} = \frac{f^2(x)}{1+x^2}$

$\frac{dy}{dx} - \left(\frac{2x}{1+x^2}\right) y = y^2$

or $\frac{1}{y^2}\frac{dy}{dx} - \left(\frac{2x}{1+x^2}\right)\frac{1}{y} = 1$

Let $-\frac{1}{y} = t \Rightarrow \frac{1}{y^2}\frac{dy}{dx} = \frac{dt}{dx}$

$\frac{dt}{dx} + \left(\frac{2x}{1+x^2}\right) t = 1$

Solution of above equation is

$-\frac{1}{y}(1+x^2) = \frac{x^3}{3} + x + c.$

Since f(0) = 1, we get c = –1

Hence, $f(x) = \frac{-3(1+x^2)}{x^3+3x-3}$.

$f\left(\frac{1}{2}\right) = \frac{30}{11}$ units.

3. Ans (2)

$P(x) - P(x+1) = 2x - 1 \ \forall x \in R$

$P(x) = ax^2 + bx + c$

$ax^2 + bx + c - [a(x+1)^2 + b(x+1) + c] = 2x - 1$

$-[ax^2 + 2ax + a + bx + b]$

$-2ax - (a + b) = 2x - 1$

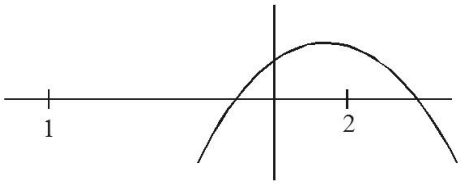
$-2a = 2 \quad a + b = 1$

$a = -1 \quad b = 2$

$P(x) = -x^2 + 2x + c$

Greatest value of

$P(x)$ in $(-\infty, 0]$ is 2



$\therefore P(0) = c = 2$

$\therefore P(x) = -x^2 + 2x + 2$

$P(1) = -1 + 2 + 2 = 3$

$P(-1) = -1 - 2 + 2 = -1$

$P(1) + P(-1) = 2$

4. Ans (1)

$$S_n = \frac{1}{n}\left(\frac{1}{\sqrt{4n^2}} + \frac{1}{\sqrt{4n^2 - 1^\alpha}} + \frac{1}{\sqrt{4n^2 - 2^\alpha}} + \ldots + \frac{1}{\sqrt{4n^2 - (n-1)^\alpha}}\right)$$

$$L = \lim_{n\to\infty}(n.S_n)$$

$$= \lim_{n\to\infty}\left(\frac{1}{\sqrt{4n^2}} + \frac{1}{\sqrt{4n^2 - 1}} + \frac{1}{\sqrt{4n^2 - 2}} + \ldots + \frac{1}{\sqrt{4n^2 - (n-1)}}\right)$$

By sandwich Theorem $L = \frac{1}{2}$

$$M = \lim_{n\to\infty}(n.S_n)$$

$$= \lim_{n\to\infty}\left(\frac{1}{\sqrt{4n^2}} + \frac{1}{\sqrt{4n^2 - 1^2}} + \frac{1}{\sqrt{4n^2 - 2^2}} + \ldots + \frac{1}{\sqrt{4n^2 - (n-1)^2}}\right)$$

$$= \lim_{n\to\infty}\sum_{r=0}^{n-1}\frac{1}{\sqrt{4n^2 - r^2}} = \int_0^1 \frac{dx}{\sqrt{4 - x^2}} = \left(\sin^{-1}\frac{x}{2}\right)_0^1$$

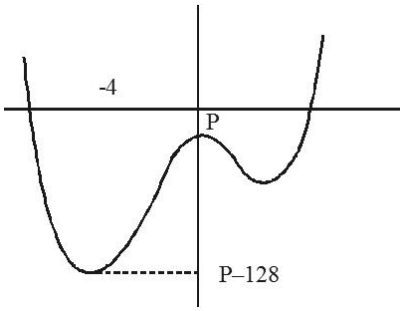
$$= \frac{\pi}{6}$$

$$\therefore L + M = \frac{1}{2} + \frac{\pi}{6} = \frac{3+\pi}{6} \Rightarrow [L + M] = 1$$

5. Ans (8)

Graph of f(x) is

so for differentiable



$P - 128 \geqslant 0$

$P \geqslant 128$

$\lambda = 128$

6. Ans (6)

$\because$ (x, 1), (y, 2), (z, 1), (2, 3) $\in A \times B$

$\Rightarrow A = \{x, y, z, 2\}$

$B = \{1, 2, 3\}$

Number of onto map = ${}^4C_2 3! = 36$

7. Ans (4)

$$\left(\left(\vec{\ell} \times \left(\vec{m} \times \vec{\ell}\right)\right) \times (\vec{n} \times \vec{m})\right) . \vec{n}$$

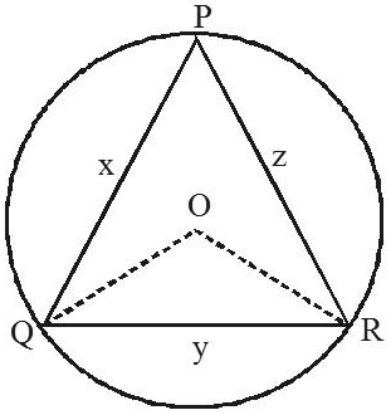
$$\left(\left(\vec{\ell} . \vec{\ell}\right) \vec{m} \times (\vec{n} \times \vec{m})\right) . \vec{n}$$

$$\left|\vec{\ell}\right|^2 \left(|\vec{m}|^2 \vec{n}\right) . \vec{n} = \left|\vec{\ell}\right|^2 |\vec{m}|^2 |\vec{n}|^2 = 36$$

$$9 \times \frac{1}{4} \times |\vec{n}|^2 = 36$$

$$|\vec{n}| = 4$$

8. Ans (1)



$$x^2 + y^2 + z^2 = 5$$

$$\angle QPR = 30$$

$$\frac{1}{2} xz . \frac{1}{2} = \frac{\sqrt{3}}{4} \Rightarrow xz = \sqrt{3}$$

$$\cos \angle QPR = \frac{x^2 + z^2 - y^2}{2xz}$$

$$\frac{\sqrt{3}}{2} = \frac{5 - 2y^2}{2 \times \sqrt{3}} \Rightarrow 2y^2 = 2 \Rightarrow y = 1$$

$$\text{from sine rule} = \frac{y}{\sin 30} = 2R \Rightarrow R = 1$$